Comprehensive Quant Interview Guide

XeLaTeX Edition

This document is a compact but rigorous interview companion. Each section contains the definitions and formulas that are most often used in quant interviews, followed by worked examples with complete solutions. The emphasis is on tools that recur in trading, research, and mathematical finance interviews: linear algebra, analysis, combinatorics, and probability.

Contents

1 Linear Algebra

Quant interviews use linear algebra for covariance matrices, regressions, factor models, optimization, and stability questions. The focus is usually on structure: what a matrix or operator can and cannot do.

1.1 Systems, Rank, Determinants, and Inverses

Definitions and formulas.

- A *permutation* of $\{1, \dots, n\}$ is a bijection σ. Its sign is $\mathrm{sgn}(\sigma) = (-1)^{\mathrm{inv}(\sigma)}$, where $\mathrm{inv}(\sigma)$ is the number of inversions. Every permutation decomposes into disjoint cycles and also into a product of transpositions. These signs are exactly the coefficients that appear in determinant expansions.
- A complex number has Cartesian and polar forms
$$z = x + iy = r(\cos\theta + i\sin\theta) = re^{i\theta}, \qquad |z| = \sqrt{x^2 + y^2}.$$
De Moivre's formula gives $(re^{i\theta})^n = r^n e^{in\theta}$, and the nth roots of unity are $e^{2\pi ik/n}$ for $k = 0, \dots, n-1$. These facts are useful whenever eigenvalues or characteristic roots are complex.
- A linear system is written $Ax = b$. It is consistent if and only if
$$\mathrm{rank}(A) = \mathrm{rank}([A \mid b]).$$
If this common rank equals the number of unknowns, the solution is unique. Otherwise,
$$x = x_p + x_h,$$
where x_p is one particular solution and x_h runs over the nullspace of A. Rank counts the number of independent constraints, so free variables appear exactly when the rank is too small.
- Elementary row operations preserve the solution set of a system, and Gaussian elimination reduces a system to echelon form. In practice this is the fastest way to detect pivots, free variables, and inconsistency.
- For an invertible square matrix,
$$A^{-1} = \frac{1}{\det(A)}\,\mathrm{adj}(A), \qquad \det(AB) = \det(A)\det(B).$$
Also, A is invertible if and only if $\det(A) \neq 0$. Invertibility means every right-hand side b has a unique solution to $Ax = b$.
- The determinant is multilinear in rows (or columns), alternates under row swaps, and measures signed volume. A zero determinant signals linear dependence of rows or columns.
- A useful rank inequality is
$$\mathrm{rank}(AB) \le \min\{\mathrm{rank}(A), \mathrm{rank}(B)\}.$$
Composition cannot create more independent directions than either map already had.
- Iterative solvers appear in numerical interviews. Writing $A = D + L + U$ gives the Jacobi iteration
$$x^{(k+1)} = D^{-1}(b - (L + U)x^{(k)}),$$

and Gauss–Seidel iteration

$$x^{(k+1)} = (D + L)^{-1}(b - Ux^{(k)}).$$

A sufficient condition for convergence is strict diagonal dominance. These methods trade exact elimination for repeated cheap updates.

Example 1.1 (Consistency and general solution)**.** Solve the system

$$\begin{aligned} x + y + z &= 1, \\ 2x + 2y + 2z &= 2, \\ x - y &= 0. \end{aligned}$$

Determine whether the solution is unique.

Solution. The second equation is just twice the first, so it contributes no new information. The third equation gives $x = y$. Substituting into the first equation yields

$$2x + z = 1.$$

Let $z = t$. Then

$$x = y = \frac{1-t}{2}, \qquad z = t.$$

Therefore the solution set is

$$\left\{ \left(\frac{1-t}{2}, \frac{1-t}{2}, t \right) : t \in \mathbb{R} \right\}.$$

This matches the rank test: $\operatorname{rank}(A) = \operatorname{rank}([A \mid b]) = 2 < 3$, so the system is consistent but has infinitely many solutions. □

1.2 Vector Spaces and Linear Maps

Definitions and formulas.

- A set V is a vector space if it is closed under addition and scalar multiplication. A *basis* is a linearly independent spanning set. The number of basis vectors is $\dim(V)$. A basis is the coordinate system intrinsic to the space.
- For subspaces $U, V \subseteq \mathbb{R}^n$,

  $$\dim(U + V) = \dim(U) + \dim(V) - \dim(U \cap V).$$

  If $U \cap V = \{0\}$, then $U \oplus V$ is a direct sum and $\dim(U \oplus V) = \dim(U) + \dim(V)$. The overlap must be subtracted because it would otherwise be counted twice.
- If $L : V \to W$ is linear, then

  $$\ker(L) = \{v \in V : L(v) = 0\}, \qquad \operatorname{im}(L) = \{L(v) : v \in V\},$$

  and the rank–nullity theorem says

  $$\dim(\ker L) + \dim(\operatorname{im} L) = \dim(V).$$

  The kernel measures directions lost by the map, while the image measures directions that survive.
- If C is the change-of-basis matrix from the new basis to the old one, then the matrix of an operator changes by similarity:

  $$A_{\text{new}} = C^{-1} A_{\text{old}} C.$$

  Similar matrices represent the same linear transformation in different coordinates.

- In regression and least squares,

$$\min_x \|Ax - b\|_2^2 \quad \Longrightarrow \quad A^T A x = A^T b.$$

The normal equations say that the residual $b - Ax$ must be orthogonal to the column space of A.

Example 1.2 (Dimension of a constrained symmetric matrix space)**.** Let subspaces $V, U \subseteq \mathbb{R}^4$ be defined by

$$V = \operatorname{span}\{v_1, v_2, v_3\}, \qquad U = \{y \in \mathbb{R}^4 : Ty = 0\},$$

where

$$v_1 = (3, -2, 2, 1)^T, \qquad v_2 = (-1, 2, -2, 3)^T, \qquad v_3 = (3, -3, -1, 2)^T,$$

and

$$T = (2, -1, -3, -2).$$

Define

$$F = \{A \in M_4(\mathbb{R}) : \text{every column of } A \text{ lies in } V,\ \text{every row of } A \text{ lies in } U\}.$$

Find

$$W = \{A \in M_4(\mathbb{R}) : A = A^T,\ A \in F\}.$$

Write an integer as the answer, or -1 if the set is not a linear space.

Solution. The set W is a linear subspace because symmetry and the row/column constraints are all linear conditions.

First compute the dimension of $V \cap U$.

- The vectors v_1, v_2, v_3 are linearly independent, so $\dim(V) = 3$.
- Since $U = \ker(T)$ with $T \neq 0$, we have $\dim(U) = 3$.
- Now

$$Tv_2 = 2(-1) - 2 + 6 - 6 = -4 \neq 0,$$

so $v_2 \notin U$. Therefore V is not contained in U, hence $V + U = \mathbb{R}^4$.

Therefore,

$$\dim(V \cap U) = \dim(V) + \dim(U) - \dim(V + U) = 3 + 3 - 4 = 2.$$

Set $I = V \cap U$, so $\dim(I) = 2$.

Now let $A \in W$. Since the columns of A lie in V and $A = A^T$, each row is the transpose of a column. Because each row lies in U, each column also lies in U. Thus every column of A lies in $I = V \cap U$. So the image of A is contained in the two-dimensional space I.

Because A is symmetric, in an orthonormal basis adapted to the decomposition

$$\mathbb{R}^4 = I \oplus I^{\perp},$$

the matrix of A has block form

$$\begin{pmatrix} B & 0 \\ 0 & 0 \end{pmatrix},$$

where B is a symmetric 2×2 real matrix. The space of symmetric 2×2 matrices has dimension

$$\frac{2(2+1)}{2} = 3.$$

Hence

$$\boxed{\dim(W) = 3}.$$

□

Example 1.3 (Least-squares solution via normal equations)**.** Find the least-squares solution of

$$Ax \approx b, \qquad A = \begin{pmatrix} 1 & 1 \\ 1 & -1 \\ 1 & 0 \end{pmatrix}, \qquad b = \begin{pmatrix} 1 \\ 0 \\ 2 \end{pmatrix}.$$

Solution. The least-squares solution satisfies the normal equations

$$A^T A x = A^T b.$$

Compute

$$A^T A = \begin{pmatrix} 3 & 0 \\ 0 & 2 \end{pmatrix}, \qquad A^T b = \begin{pmatrix} 3 \\ 1 \end{pmatrix}.$$

Therefore

$$\begin{pmatrix} 3 & 0 \\ 0 & 2 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 3 \\ 1 \end{pmatrix},$$

so

$$x_1 = 1, \qquad x_2 = \frac{1}{2}.$$

Hence the least-squares solution is

$$\boxed{x^* = \begin{pmatrix} 1 \\ \frac{1}{2} \end{pmatrix}}.$$

Indeed,

$$Ax^* = \begin{pmatrix} 3/2 \\ 1/2 \\ 1 \end{pmatrix}, \qquad r = b - Ax^* = \begin{pmatrix} -1/2 \\ -1/2 \\ 1 \end{pmatrix},$$

and one checks that r is orthogonal to both columns of A, as it should be. □

Example 1.4 (Prescribing an eigenvector and excluding a vector from the image)**.** Let $v, w \in \mathbb{R}^n$ be linearly independent. Does there always exist an $n \times n$ matrix A such that v is an eigenvector of A with eigenvalue 5, while $w \notin \operatorname{im}(A)$?

Solution. Yes, always. Since v, w are linearly independent, extend $\{v, w\}$ to a basis $\{v, w, e_3, \dots, e_n\}$ of $\mathbb{R}^n$. A linear map is determined by its values on a basis, so define A by

$$Av = 5v, \qquad Aw = 0, \qquad Ae_3 = \cdots = Ae_n = 0.$$

By construction v is an eigenvector of A with eigenvalue 5. Moreover

$$\operatorname{im}(A) = \operatorname{span}\{5v, 0, \dots, 0\} = \operatorname{span}\{v\},$$

and $w \notin \operatorname{span}\{v\}$ precisely because v, w are linearly independent. Hence

$$\boxed{w \notin \operatorname{im}(A)}$$

for this choice of A. Concretely: writing any $x \in \mathbb{R}^n$ in basis coordinates $x = \alpha v + \beta w + \sum_i \gamma_i e_i$, one has $Ax = 5\alpha v$. □

Example 1.5 (Equal rank from an invertibility hypothesis)**.** Let A, B be $n \times n$ matrices with $A^2 = A$, $B^2 = B$, and suppose $I - (A + B)$ is invertible. Prove that $\operatorname{rank} A = \operatorname{rank} B$.

Solution. Let $C = I - A - B$, invertible by hypothesis. Since A is idempotent, every $v \in \operatorname{im}(A)$ satisfies $Av = v$ (write $v = Au$; then $Av = A^2u = Au = v$), and likewise $Bv = v$ for $v \in \operatorname{im}(B)$.

Take $v \in \operatorname{im}(A)$. Then

$$Cv = v - Av - Bv = v - v - Bv = -Bv \in \operatorname{im}(B),$$

so C maps $\operatorname{im}(A)$ into $\operatorname{im}(B)$. As C is invertible it is injective, so its restriction to $\operatorname{im}(A)$ is an injective map into $\operatorname{im}(B)$, giving

$$\operatorname{rank} A = \dim \operatorname{im}(A) \leq \dim \operatorname{im}(B) = \operatorname{rank} B.$$

Symmetrically, for $v \in \operatorname{im}(B)$,

$$Cv = v - Av - Bv = v - Av - v = -Av \in \operatorname{im}(A),$$

so the same argument gives $\operatorname{rank} B \leq \operatorname{rank} A$. Combining,

$$\boxed{\operatorname{rank} A = \operatorname{rank} B}.$$

□

1.3 Eigenvalues, Jordan Form, and Singular Value Decomposition

Definitions and formulas.

- A scalar λ is an eigenvalue of A if there exists $v \neq 0$ such that $Av = \lambda v$. The eigenspace is $E_\lambda = \ker(A - \lambda I)$. Eigenvectors are directions whose orientation is preserved by the transformation.
- The characteristic polynomial is $p_A(\lambda) = \det(\lambda I - A)$. The Hamilton–Cayley theorem states that $p_A(A) = 0$. This lets you reduce high powers of a matrix to lower ones.
- For an $n \times n$ matrix, the sum of eigenvalues counted with algebraic multiplicity is $\operatorname{tr}(A)$ and the product is $\det(A)$. These are quick consistency checks when you compute a spectrum.
- A matrix is diagonalizable if and only if for every eigenvalue, algebraic multiplicity equals geometric multiplicity. Equivalently, the space has a full basis of eigenvectors.
- In particular, if the characteristic polynomial splits and has distinct roots, the matrix is automatically diagonalizable. Distinct eigenvalues automatically give linearly independent eigenvectors.
- A nilpotent operator satisfies $N^k = 0$ for some k. Its Jordan form is a direct sum of Jordan blocks with zero diagonal. Nilpotent parts measure failure of diagonalizability.
- If N is nilpotent on an n-dimensional space, then its only eigenvalue is 0 and its characteristic polynomial is λ^n. Repeated application eventually kills every vector.
- The singular value decomposition is

  $$A = U\Sigma V^T,$$

  where U, V are orthogonal and Σ is diagonal with nonnegative entries $\sigma_i = \sqrt{\lambda_i(A^TA)}$. The SVD separates input directions, stretching factors, and output directions.
- The singular values measure the principal stretching factors of the linear map $x \mapsto Ax$. The rank of A equals the number of nonzero singular values. For rectangular matrices they play the role that absolute eigenvalues play for symmetric matrices.

- If A is real symmetric, then the spectral theorem gives

$$A = Q\Lambda Q^T,$$

where Q is orthogonal and Λ is diagonal with real eigenvalues. Symmetric matrices therefore act like independent scalar multipliers along orthogonal directions.

- The Frobenius norm is

$$\|A\|_F = \sqrt{\operatorname{tr}(A^T A)} = \left(\sum_{i,j} a_{ij}^2\right)^{1/2} = \left(\sum_i \sigma_i^2\right)^{1/2}.$$

It is the Euclidean norm of the matrix entries viewed as one long vector, and it is the natural norm for entrywise least-squares error.

- The best rank-k approximation in Frobenius norm is obtained by truncating the SVD:

$$A_k = \sum_{i=1}^{k} \sigma_i u_i v_i^T.$$

Keeping the largest singular values preserves the most important directions of variation.

Example 1.6 (Using Hamilton–Cayley to compute powers)**.** Let

$$A = \begin{pmatrix} 2 & 1 \\ 0 & 2 \end{pmatrix}.$$

Find a formula for A^n.

Solution. Write

$$A = 2I + N, \qquad N = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \qquad N^2 = 0.$$

Because $N^2 = 0$, the binomial expansion truncates:

$$A^n = (2I + N)^n = \sum_{k=0}^{n} \binom{n}{k} 2^{n-k} N^k = 2^n I + n2^{n-1} N.$$

Hence

$$A^n = \begin{pmatrix} 2^n & n2^{n-1} \\ 0 & 2^n \end{pmatrix}.$$

This is exactly what Hamilton–Cayley predicts, since the characteristic polynomial is $(\lambda - 2)^2$ and therefore $(A - 2I)^2 = 0$. □

Example 1.7 (Singular values of a rank-one matrix)**.** Let

$$A = \begin{pmatrix} 3 & 0 \\ 4 & 0 \end{pmatrix}.$$

Find the singular values of A and write down one SVD.

Solution. Compute

$$A^T A = \begin{pmatrix} 25 & 0 \\ 0 & 0 \end{pmatrix}.$$

Its eigenvalues are 25 and 0, so the singular values are

$$\sigma_1 = 5, \qquad \sigma_2 = 0.$$

Take

$$v_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \qquad v_2 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}.$$

Then

$$u_1 = \frac{Av_1}{\|Av_1\|} = \frac{1}{5}\begin{pmatrix} 3 \\ 4 \end{pmatrix} = \begin{pmatrix} 3/5 \\ 4/5 \end{pmatrix}.$$

Choose an orthonormal vector perpendicular to u_1, for example

$$u_2 = \begin{pmatrix} -4/5 \\ 3/5 \end{pmatrix}.$$

Therefore one valid SVD is

$$U = \begin{pmatrix} 3/5 & -4/5 \\ 4/5 & 3/5 \end{pmatrix}, \qquad \Sigma = \begin{pmatrix} 5 & 0 \\ 0 & 0 \end{pmatrix}, \qquad V = I.$$

So

$$\boxed{A = U\Sigma V^T}$$

with singular values 5 and 0. □

Example 1.8 (Centralizer dimension for cyclic vs. non-cyclic matrices)**.** For an $n \times n$ real matrix A and $v \in \mathbb{R}^n$, let $U(A) = \{X \in \mathrm{Mat}_n(\mathbb{R}) : AX = XA\}$ and $W(A, v) = \mathrm{span}\{v, Av, A^2v, \dots\}$ (the A-cyclic subspace generated by v). (a) If $\dim W(A, v) = n$ for *every* $v \neq 0$, what is the largest possible $\dim U(A)$? (b) If $\dim W(A, v) < n$ for *every* v, what is the smallest possible $\dim U(A)$?

Solution. **Background fact.** For any A, $\dim U(A) \geq n$, with equality iff A is *non-derogatory* (its minimal polynomial has degree n, i.e. equals the characteristic polynomial). More precisely, if the Jordan blocks for a single eigenvalue have sizes $n_1 \geq n_2 \geq \cdots \geq n_r$ (a partition of that eigenvalue's multiplicity), that eigenvalue contributes $\sum_i (2i-1)n_i$ to $\dim U(A)$; summing over eigenvalues and minimizing $\sum(2i-1)n_i$ for fixed $\sum n_i$ shows single blocks ($r = 1$) are optimal, giving the bound $\dim U(A) \geq n$.

(**a**) If v_0 is any vector with $\dim W(A, v_0) = n$, then $v_0, Av_0, \dots, A^{n-1}v_0$ are linearly independent, so the minimal polynomial of A has degree $\geq n$, hence exactly n: A is non-derogatory. By the background fact, $\dim U(A) = n$ exactly. So under the hypothesis of (a) (which certainly implies *some* cyclic vector exists), the dimension is forced to be

$$\boxed{n}.$$

(Such A exist for even n: e.g. for $n = 2$, $A = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$ has no real eigenvalues, so $\{v, Av\}$ is dependent only if v is a real eigenvector — impossible — hence every nonzero v is cyclic.)

(**b**) Now no vector is cyclic, so A is derogatory: some eigenvalue has $r \geq 2$ Jordan blocks. To minimize $\dim U(A) = \sum(2i-1)n_i$ (summed over all eigenvalues) subject to this, isolate the derogatory eigenvalue's contribution $n_1 + 3n_2$ with $n_1 \geq n_2 \geq 1$: writing the eigenvalue's multiplicity as $m = n_1 + n_2$, this equals $m + 2n_2 \geq m + 2$, minimized at $n_2 = 1$ (a single "stray" Jordan block of size 1 split off from an otherwise full block of size $n-1$). Every other eigenvalue, kept non-derogatory, contributes exactly its own multiplicity. Summing, the total is minimized at

$$\boxed{n+2},$$

attained by a single eigenvalue with Jordan blocks of sizes $(n-1, 1)$. □

1.4 Bilinear Forms, Euclidean Spaces, and Hermitian Structure

Definitions and formulas.

- A bilinear form on $\mathbb{R}^n$ is a map $B(x, y) = x^T Ay$ that is linear in each argument. If $A = A^T$, then the associated quadratic form is $Q(x) = x^T Ax$. The quadratic form records what the bilinear form does when both inputs are the same vector.
- Over $\mathbb{R}$, a symmetric bilinear form and its quadratic form determine each other through the polarization identity
$$B(x, y) = \frac{1}{2}(Q(x + y) - Q(x) - Q(y)).$$
This is the precise connection between bilinear and quadratic viewpoints.
- Under a change of basis with matrix C, the matrix of a bilinear form transforms by congruence:
$$A_{\text{new}} = C^T A_{\text{old}} C.$$
Congruence is the correct notion because both inputs of the form are being re-expressed in the new coordinates.
- A symmetric matrix is positive definite if $x^T Ax > 0$ for all $x \neq 0$. Sylvester's criterion says this holds if and only if all leading principal minors are positive. Positive definiteness means the quadratic form behaves like a genuine squared length.
- In Euclidean space,
$$|\langle u, v\rangle| \leq \|u\| \, \|v\|,$$
and if $G = (\langle v_i, v_j\rangle)$ is the Gram matrix, then the squared volume of the parallelotope spanned by $v_1, \dots, v_k$ is $\det(G)$. The Gram matrix packages all pairwise inner products at once.
- Gram–Schmidt orthogonalization turns a basis into an orthogonal basis. Orthogonal projections onto $\operatorname{span}\{q_1, \dots, q_k\}$ are computed by
$$P = Q(Q^T Q)^{-1} Q^T,$$
where $Q = [q_1 \ \cdots \ q_k]$. Orthogonal coordinates simplify both geometry and computation.
- In complex vector spaces, Hermitian matrices satisfy $A = A^*$ and have real eigenvalues with orthonormal eigenvectors. They are the complex analog of real symmetric matrices.

Example 1.9 (Why $A^T A$ is always positive semidefinite)**.** Prove that for every real matrix A, the matrix $A^T A$ is positive semidefinite.

Solution. Take any vector $x \in \mathbb{R}^n$. Then
$$x^T A^T Ax = (Ax)^T (Ax) = \|Ax\|_2^2 \geq 0.$$
Therefore $x^T A^T Ax \geq 0$ for all x, which is exactly the definition of positive semidefinite. Equality holds if and only if $Ax = 0$. □

Example 1.10 (A bilinear-form matrix that is similar to an idempotent)**.** Let $Q : V \times W \to \mathbb{R}$ be a bilinear map. In bases α of V and β of W, write
$$Q(v, w) = v_\alpha^T Q_{\alpha,\beta} w_\beta.$$
Now regard $Q_{\alpha,\beta}$ as the matrix of a linear operator on a 4-dimensional real space. Suppose that in some other basis this operator has matrix B satisfying
$$B^2 = B.$$

It is also known that in some pair of bases ω, γ for the bilinear map,

$$Q_{\omega,\gamma} = \begin{pmatrix} 2 & 0 & 2 & 3 \\ 0 & 0 & 0 & 2 \\ 4 & 0 & 4 & 5 \\ 2 & 0 & 2 & 2 \end{pmatrix}.$$

Find the maximum and minimum possible dimensions of an eigenspace of this operator.

Solution. For a bilinear map, changing the basis in V and the basis in W acts by left and right multiplication with invertible matrices:

$$Q_{\omega,\gamma} = C^T Q_{\alpha,\beta} D.$$

Therefore the rank of the matrix is invariant under such changes of bases.

Now inspect the given matrix

$$M = \begin{pmatrix} 2 & 0 & 2 & 3 \\ 0 & 0 & 0 & 2 \\ 4 & 0 & 4 & 5 \\ 2 & 0 & 2 & 2 \end{pmatrix}.$$

Its second column is zero and its third column equals its first column, so $\operatorname{rank}(M) \leq 2$. On the other hand, the first and fourth columns are linearly independent, so $\operatorname{rank}(M) = 2$. Hence

$$\operatorname{rank}(Q_{\alpha,\beta}) = 2.$$

Now switch to the operator point of view. The matrix $Q_{\alpha,\beta}$ is similar to B, and similarity preserves rank, so

$$\operatorname{rank}(B) = 2.$$

Because $B^2 = B$, the operator is idempotent. For any idempotent operator T,

$$\operatorname{im}(T) = E_1(T), \qquad \ker(T) = E_0(T).$$

Indeed, if $y = Tx$, then $Ty = T^2x = Tx = y$, so every vector in the image is a 1-eigenvector. Conversely, if $Ty = y$, then $y = Tx$ with $x = y$, so $y \in \operatorname{im}(T)$. The kernel is exactly the eigenspace for eigenvalue 0.

Therefore for B we get

$$\dim E_1(B) = \operatorname{rank}(B) = 2, \qquad \dim E_0(B) = 4 - \operatorname{rank}(B) = 2.$$

So both eigenspaces have the same dimension. Hence the maximum and minimum possible dimensions of an eigenspace are both

$$\boxed{2}.$$

Equivalently, the only possible answer is

$$\boxed{\min = \max = 2}.$$

□

Example 1.11 (Matrix-valued quadratic form induced by a positive definite matrix)**.** Let $A \in \mathbb{R}^{n\times n}$ be symmetric positive definite. Define

$$q(X) = \operatorname{tr}(X^T AX), \qquad X \in \mathbb{R}^{n\times m}.$$

Is q a positive definite quadratic form on the vector space $\mathbb{R}^{n\times m}$?

Solution. Yes. The expression is quadratic because for any scalar c,

$$q(cX) = \operatorname{tr}((cX)^T A(cX)) = c^2 \operatorname{tr}(X^T AX) = c^2 q(X).$$

The associated bilinear form is

$$B(X, Y) = \operatorname{tr}(X^T AY),$$

since, using the symmetry of A,

$$q(X + Y) = q(X) + q(Y) + 2 \operatorname{tr}(X^T AY).$$

To check positive definiteness, write the columns of X as

$$X = [x_1 \ \cdots \ x_m].$$

Then

$$q(X) = \operatorname{tr}(X^T AX) = \sum_{j=1}^{m} x_j^T Ax_j.$$

Because A is positive definite, each term satisfies $x_j^T Ax_j \geq 0$, and it is strictly positive whenever $x_j \neq 0$. Therefore if $X \neq 0$, at least one column x_j is nonzero, so

$$q(X) > 0.$$

Hence q is a positive definite quadratic form on $\mathbb{R}^{n\times m}$.

There is also a useful norm interpretation. Since A is positive definite, it has a symmetric square root $A^{1/2}$, and

$$q(X) = \operatorname{tr}(X^T A^{1/2} A^{1/2} X) = \operatorname{tr}((A^{1/2}X)^T (A^{1/2}X)) = \|A^{1/2}X\|_F^2.$$

So q is literally the squared Frobenius norm after the linear change of variables $X \mapsto A^{1/2}X$. □

Example 1.12 (When do two matrices represent the same bilinear form?)**.** For which $a \in \mathbb{R}$ can

$$A = \begin{pmatrix} 1 & 4 - a - a^2 \\ 2 & -1 \end{pmatrix}, \qquad B = \begin{pmatrix} -a - 1 & 3 \\ 3 & -5 \end{pmatrix}$$

be matrices of the *same* bilinear form $V \times V \to \mathbb{R}$ in two different bases?

Solution. Two matrices represent the same bilinear form in different bases iff they are *congruent*: $B = C^T AC$ for some invertible C. Write $A = S + K$ as symmetric plus antisymmetric parts; since $(C^T AC)^T$-splitting commutes with congruence, $C^T AC = C^T SC + C^T KC$ is again a symmetric-plus-antisymmetric split, so congruence acts separately on S and K — in particular it preserves $\operatorname{rank} K$ (for a 2×2 antisymmetric matrix, $K = 0$ or $\operatorname{rank} K = 2$).

Here B is symmetric for every a, so $K_B = 0$, forcing $K_A = 0$ too: A must be symmetric, i.e. $4 - a - a^2 = 2$, i.e. $a^2 + a - 2 = (a - 1)(a + 2) = 0$, so $a \in \{-2, 1\}$. In both cases $A = \begin{pmatrix} 1 & 2 \\ 2 & -1 \end{pmatrix}$ (the same matrix, since $4 - a - a^2 = 2$ at both roots): $\det A = -5 < 0$, so A has signature $(1, 1)$.

By Sylvester's law of inertia, two real symmetric matrices are congruent iff they have the same rank and signature, so we just need B's signature at each candidate:

- $a = -2$: $B = \begin{pmatrix} 1 & 3 \\ 3 & -5 \end{pmatrix}$, $\det B = -5 - 9 = -14 < 0$, signature $(1, 1)$ — matches A.
- $a = 1$: $B = \begin{pmatrix} -2 & 3 \\ 3 & -5 \end{pmatrix}$, $\det B = 10 - 9 = 1 > 0$ with $\operatorname{tr} B = -7 < 0$, so both eigenvalues are negative: signature $(0, 2)$ — does *not* match A.

So only $a = -2$ works:

$$\boxed{a = -2}.$$

□

2 Calculus and Real Analysis

Interview analysis questions often look innocent but test whether you can move between definitions, expansions, and clean arguments. The core techniques are asymptotics, derivatives, optimization, integrals, and series.

2.1 Sequences, Limits, Continuity, and Asymptotics

Definitions and formulas.

- A sequence (a_n) converges to L if for every $\varepsilon > 0$ there exists N such that $|a_n - L| < \varepsilon$ for all $n \geq N$.
- A bounded monotone sequence converges. Every bounded sequence has a convergent subsequence (Bolzano–Weierstrass).
- Important limits:

$$\lim_{x\to 0} \frac{\sin x}{x} = 1, \qquad \lim_{x\to 0} \frac{1-\cos x}{x^2} = \frac{1}{2}, \qquad \lim_{x\to\infty} \left(1 + \frac{1}{x}\right)^x = e.$$

- A function is continuous at a if $\lim_{x\to a} f(x) = f(a)$. Continuous functions on compact intervals are bounded and attain maxima and minima.
- Asymptotic notation:

$$f(x) = O(g(x)) \iff \frac{f(x)}{g(x)} \text{ is bounded}, \qquad f(x) = o(g(x)) \iff \frac{f(x)}{g(x)} \to 0.$$

- Taylor expansions near 0 that should be memorized:

$$\sin x = x - \frac{x^3}{6} + O(x^5), \qquad \cos x = 1 - \frac{x^2}{2} + O(x^4), \qquad \log(1+x) = x - \frac{x^2}{2} + O(x^3).$$

Example 2.1 (A standard small-x limit)**.** Compute

$$\lim_{x\to 0} \frac{\sin x - x}{x^3}.$$

Solution. Use the Taylor expansion

$$\sin x = x - \frac{x^3}{6} + O(x^5).$$

Then

$$\sin x - x = -\frac{x^3}{6} + O(x^5),$$

so

$$\frac{\sin x - x}{x^3} = -\frac{1}{6} + O(x^2).$$

Therefore

$$\boxed{\lim_{x\to 0} \frac{\sin x - x}{x^3} = -\frac{1}{6}}.$$

□

Example 2.2 (Chaining two equivalent infinitesimals)**.** Suppose $\lim_{x\to 0} f(x)/\sin x = 2$. Find $\lim_{x\to 0} \ln(1+3x)/f(x)$.

Solution. Write the target limit as a product of three limits that each exist:

$$\frac{\ln(1+3x)}{f(x)} = \frac{\ln(1+3x)}{x} \cdot \frac{x}{\sin x} \cdot \frac{\sin x}{f(x)}.$$

As $x \to 0$: $\ln(1+3x)/x \to 3$, $x/\sin x \to 1$, and $\sin x/f(x) \to 1/2$ (the reciprocal of the given limit, valid since $f(x)/\sin x \to 2 \neq 0$ forces $f(x) \neq 0$ near 0). Hence

$$\boxed{\lim_{x\to 0} \frac{\ln(1+3x)}{f(x)} = 3 \cdot 1 \cdot \frac{1}{2} = \frac{3}{2}}.$$

□

2.2 Trigonometric and Hyperbolic Identities for Reference

Useful identities.

- Pythagorean identities:

$$\sin^2 x + \cos^2 x = 1, \qquad 1 + \tan^2 x = \sec^2 x, \qquad 1 + \cot^2 x = \csc^2 x.$$

- Angle-sum formulas:

$$\sin(a \pm b) = \sin a \cos b \pm \cos a \sin b,$$
$$\cos(a \pm b) = \cos a \cos b \mp \sin a \sin b.$$

- Double-angle and half-angle formulas:

$$\sin 2x = 2 \sin x \cos x,$$
$$\cos 2x = \cos^2 x - \sin^2 x = 1 - 2\sin^2 x = 2\cos^2 x - 1,$$
$$\sin^2 \frac{x}{2} = \frac{1 - \cos x}{2}, \qquad \cos^2 \frac{x}{2} = \frac{1 + \cos x}{2}.$$

- Product-to-sum formulas:

$$\sin a \sin b = \frac{1}{2}(\cos(a-b) - \cos(a+b)),$$
$$\cos a \cos b = \frac{1}{2}(\cos(a-b) + \cos(a+b)),$$
$$\sin a \cos b = \frac{1}{2}(\sin(a+b) + \sin(a-b)).$$

- Hyperbolic identities:

$$\sinh t = \frac{e^t - e^{-t}}{2}, \qquad \cosh t = \frac{e^t + e^{-t}}{2}, \qquad \cosh^2 t - \sinh^2 t = 1.$$

- Derivatives of hyperbolic functions:

$$\frac{d}{dt} \sinh t = \cosh t, \qquad \frac{d}{dt} \cosh t = \sinh t.$$

- Useful substitution pattern:

$$x = e^t \quad \Longrightarrow \quad x - \frac{1}{x} = 2 \sinh t.$$

  In some texts, especially older or Eastern European ones, sinh is denoted by sh and cosh by ch.

Example 2.3 (A power-reduction integral)**.** Evaluate

$$\int_0^{\pi/2} \sin^4 x \, dx.$$

Solution. Use the half-angle identity

$$\sin^2 x = \frac{1 - \cos 2x}{2}.$$

Then

$$\sin^4 x = \left(\frac{1 - \cos 2x}{2} \right)^2 = \frac{1}{4} \left(1 - 2\cos 2x + \cos^2 2x \right).$$

Since

$$\cos^2 2x = \frac{1 + \cos 4x}{2},$$

we get

$$\sin^4 x = \frac{3}{8} - \frac{1}{2} \cos 2x + \frac{1}{8} \cos 4x.$$

Therefore

$$\int_0^{\pi/2} \sin^4 x \, dx = \int_0^{\pi/2} \left(\frac{3}{8} - \frac{1}{2} \cos 2x + \frac{1}{8} \cos 4x \right) dx.$$

The cosine terms integrate to 0 on $[0, \pi/2]$, so

$$\int_0^{\pi/2} \sin^4 x \, dx = \frac{3}{8} \cdot \frac{\pi}{2} = \frac{3\pi}{16}.$$

Hence

$$\boxed{\int_0^{\pi/2} \sin^4 x \, dx = \frac{3\pi}{16}}.$$

□

2.3 Standard Substitutions for Reference

Useful substitution patterns.

- For integrands involving $1 + x^2$ or rational functions of x and $1 + x^2$, use

  $$x = \tan t, \qquad dx = \sec^2 t \, dt, \qquad 1 + x^2 = \sec^2 t.$$

  This turns expressions involving $1 + x^2$ into trigonometric squares and is especially useful for integrals such as

  $$\int \frac{dx}{1 + x^2}, \qquad \int \frac{x^2}{(1 + x^2)^2} \, dx.$$

- For radicals of the form $\sqrt{1 + x^2}$ or, more generally, $\sqrt{a^2 + x^2}$, a hyperbolic substitution is often cleaner than a trigonometric one:

  $$x = \sinh t, \qquad dx = \cosh t \, dt, \qquad \sqrt{1 + x^2} = \cosh t.$$

  More generally,

  $$x = a \sinh t, \qquad dx = a \cosh t \, dt, \qquad \sqrt{a^2 + x^2} = a \cosh t.$$

  This is useful because $\cosh^2 t - \sinh^2 t = 1$ has the same algebraic role as $1 + \sinh^2 t = \cosh^2 t$.

- For radicals of the form $\sqrt{a^2 - x^2}$, the standard circular substitution is
$$x = a \sin t, \qquad dx = a \cos t \, dt, \qquad \sqrt{a^2 - x^2} = a \cos t.$$
For radicals of the form $\sqrt{x^2 - a^2}$, useful choices are
$$x = a \sec t \quad \text{or} \quad x = a \cosh t.$$
- For rational functions of $\sin x$ and $\cos x$, use the tangent half-angle substitution
$$u = \tan \frac{x}{2}.$$
Then
$$\sin x = \frac{2u}{1+u^2}, \qquad \cos x = \frac{1-u^2}{1+u^2}, \qquad dx = \frac{2}{1+u^2}\, du.$$
This converts any rational expression in $\sin x$ and $\cos x$ into a rational function of u.
- A useful decision rule is: if the integrand contains a quadratic expression under a square root, choose a substitution that turns that quadratic into a square; if the integrand is rational in $\sin x$ and $\cos x$, use $u = \tan(x/2)$; if exponentials create expressions like $x - 1/x$, try $x = e^t$ and rewrite the result in terms of $\sinh t$ or $\cosh t$.

Example 2.4 (Hyperbolic substitution in a definite integral)**.** Evaluate
$$\int_0^1 \sqrt{1+x^2}\, dx.$$

Solution. Because the integrand contains $\sqrt{1+x^2}$, use the substitution
$$x = \sinh t, \qquad dx = \cosh t \, dt, \qquad \sqrt{1+x^2} = \cosh t.$$
When $x = 0$, we have $t = 0$. When $x = 1$, we have $t = \operatorname{arsinh}(1)$.

Thus
$$\int_0^1 \sqrt{1+x^2}\, dx = \int_0^{\operatorname{arsinh}(1)} \cosh^2 t \, dt.$$
Using
$$\cosh^2 t = \frac{1 + \cosh 2t}{2},$$
we obtain
$$\int \cosh^2 t \, dt = \frac{1}{2}(t + \sinh t \cosh t).$$
Therefore
$$\int_0^1 \sqrt{1+x^2}\, dx = \frac{1}{2}(t + \sinh t \cosh t)\Big|_0^{\operatorname{arsinh}(1)}.$$
At $t = \operatorname{arsinh}(1)$, we have $\sinh t = 1$ and $\cosh t = \sqrt{2}$. Hence
$$\int_0^1 \sqrt{1+x^2}\, dx = \frac{1}{2}(\operatorname{arsinh}(1) + \sqrt{2}).$$
Since
$$\operatorname{arsinh}(1) = \log(1 + \sqrt{2}),$$
the final answer is
$$\boxed{\int_0^1 \sqrt{1+x^2}\, dx = \frac{\sqrt{2} + \log(1+\sqrt{2})}{2}}.$$
□

2.4 One-Variable Differential Calculus

Definitions and formulas.

- If f is differentiable at x, then

$$f'(x) = \lim_{h\to 0}\frac{f(x+h)-f(x)}{h}.$$

  Differentiability implies continuity.

- Rolle's theorem: if $f(a) = f(b)$ and f is continuous on $[a,b]$ and differentiable on (a,b), then $f'(c) = 0$ for some $c \in (a,b)$.

- Lagrange's mean value theorem:

$$f'(c) = \frac{f(b)-f(a)}{b-a}$$

  for some $c \in (a,b)$.

- Taylor's formula around a is

$$f(x) = \sum_{k=0}^{n}\frac{f^{(k)}(a)}{k!}(x-a)^k + R_n(x),$$

  with Lagrange remainder

$$R_n(x) = \frac{f^{(n+1)}(\xi)}{(n+1)!}(x-a)^{n+1}$$

  for some ξ between a and x.

- Critical point tests: if $f'(x_0) = 0$ and $f''(x_0) > 0$, then x_0 is a local minimum; if $f''(x_0) < 0$, it is a local maximum.

- L'Hopital's rule handles $0/0$ and ∞/∞ forms when the relevant hypotheses are satisfied.

Example 2.5 (Optimization on $(0,\infty)$)**.** Find the global minimum of

$$f(x) = x - \log x, \qquad x > 0.$$

Solution. Differentiate:

$$f'(x) = 1 - \frac{1}{x} = \frac{x-1}{x}.$$

The only critical point is $x = 1$. The second derivative is

$$f''(x) = \frac{1}{x^2} > 0,$$

so $x = 1$ is a strict local minimum. Since $f(x) \to \infty$ as $x \downarrow 0$ and also $f(x) \to \infty$ as $x \to \infty$, this local minimum is global. The minimum value is

$$f(1) = 1.$$

□

2.5 Multivariable Calculus and Constrained Optimization

Definitions and formulas.

- For $f : \mathbb{R}^n \to \mathbb{R}$, the gradient is

  $$\nabla f = \left(\frac{\partial f}{\partial x_1}, \dots, \frac{\partial f}{\partial x_n} \right),$$

  and the directional derivative along a unit vector u is $D_u f = \nabla f \cdot u$.
- The Hessian matrix is

  $$H_f = \left(\frac{\partial^2 f}{\partial x_i \partial x_j} \right)_{i,j}.$$

  At a critical point, a positive definite Hessian implies a local minimum; a negative definite Hessian implies a local maximum.
- To optimize $f(x)$ subject to $g(x) = 0$, introduce the Lagrangian

  $$\mathcal{L}(x, \lambda) = f(x) - \lambda g(x)$$

  and solve $\nabla_x \mathcal{L} = 0$ together with $g(x) = 0$.
- In optimization algorithms, gradient descent updates

  $$x_{k+1} = x_k - \eta \nabla f(x_k).$$

- The implicit function theorem says that if $F(x, y) = 0$ and $F_y(x_0, y_0) \neq 0$, then locally one can write y as a differentiable function of x.

Example 2.6 (Lagrange multipliers)**.** Minimize

$$f(x, y) = x^2 + y^2$$

subject to the linear constraint

$$x + 2y = 1.$$

Solution. Set

$$\mathcal{L}(x, y, \lambda) = x^2 + y^2 - \lambda(x + 2y - 1).$$

Then

$$2x - \lambda = 0, \qquad 2y - 2\lambda = 0, \qquad x + 2y = 1.$$

From the first two equations,

$$x = \frac{\lambda}{2}, \qquad y = \lambda.$$

Substitute into the constraint:

$$\frac{\lambda}{2} + 2\lambda = 1 \quad \Longrightarrow \quad \frac{5\lambda}{2} = 1 \quad \Longrightarrow \quad \lambda = \frac{2}{5}.$$

Hence

$$x = \frac{1}{5}, \qquad y = \frac{2}{5}.$$

The minimum value is

$$f\left(\frac{1}{5}, \frac{2}{5} \right) = \frac{1}{25} + \frac{4}{25} = \frac{1}{5}.$$

Because the objective is strictly convex and the constraint is affine, this minimizer is unique. □

Example 2.7 (An extremal problem with infimum zero)**.** Let M be the set of continuous, non-increasing functions $f : [0,1] \to \mathbb{R}$ with $f(1) = 0$ and $f \not\equiv 0$. Find

$$\inf_{f \in M} \sup_{x \in [0,1]} \frac{xf(x)}{\int_0^1 f(t)\,dt}.$$

Solution. Since f is non-increasing with $f(1) = 0$, automatically $f \geq 0$ on $[0,1]$. Under $f \mapsto cf$ ($c > 0$) both the numerator's supremum and the denominator scale by c, so the ratio is scale-invariant; we may search among a convenient family.

For an integer $n \geq 2$, define

$$f_n(x) = \min\left(n, \frac{1}{x}\right) - 1 \quad (x > 0), \qquad f_n(0) := n - 1,$$

i.e. $f_n(x) = n-1$ on $[0, 1/n]$ and $f_n(x) = \frac{1}{x} - 1$ on $[1/n, 1]$. This is continuous and non-increasing (the two pieces agree and are each non-increasing), with $f_n(1) = 0$, so $f_n \in M$.

Numerator. On $[0, 1/n]$, $xf_n(x) = x(n-1)$ increases to $\frac{n-1}{n}$ at $x = 1/n$. On $[1/n, 1]$, $xf_n(x) = x\left(\frac{1}{x} - 1\right) = 1 - x$ decreases from the same value $\frac{n-1}{n}$. Hence

$$\sup_x xf_n(x) = \frac{n-1}{n} < 1.$$

Denominator.

$$\int_0^1 f_n = \int_0^{1/n} (n-1)\,dt + \int_{1/n}^1 \left(\frac{1}{t} - 1\right) dt = \frac{n-1}{n} + \Big[\ln t - t\Big]_{1/n}^1 = \frac{n-1}{n} + \left(\ln n + \frac{1}{n} - 1\right) = \ln n.$$

Ratio.

$$\frac{\sup_x xf_n(x)}{\int_0^1 f_n} = \frac{(n-1)/n}{\ln n} \xrightarrow[n\to\infty]{} 0,$$

since the numerator stays bounded below 1 while the denominator grows without bound. The ratio is always strictly positive for $f \in M$, but by taking n large it can be made smaller than any $\varepsilon > 0$, so the infimum is not attained and equals

$$\boxed{0}.$$

□

Example 2.8 (Reading a gradient field: degenerate directions and gradient descent)**.** A plotted gradient field of a smooth $f(x,y)$ shows a periodic pattern: near the vertical grid lines the field is essentially vertical, near the horizontal grid lines it is essentially horizontal, and the pattern is symmetric under swapping the roles of x and y and under reflections — the signature of a field built from a product of two matching one-dimensional oscillations, $f(x,y) = g(x)g(y)$ with g satisfying $g'' = -c^2 g$ for some constant c (i.e. g is a sinusoid). (a) Along which lines is the Hessian of f singular, and how many families are there? (b) Is it possible to have $\nabla f \neq 0$ at a point from which gradient descent never reaches a minimum of f?

Solution. **(a)** Take $f(x,y) = g(x)g(y)$ with $g'' = -c^2 g$ (so g is, up to scale and phase, $\cos(cx)$). Then

$$f_{xx} = g''(x)g(y) = -c^2 f, \qquad f_{yy} = g(x)g''(y) = -c^2 f, \qquad f_{xy} = g'(x)g'(y).$$

So

$$\begin{aligned} \det H = f_{xx}f_{yy} - f_{xy}^2 &= c^4 g(x)^2 g(y)^2 - g'(x)^2 g'(y)^2 \\ &= (c^2 g(x)g(y) - g'(x)g'(y))(c^2 g(x)g(y) + g'(x)g'(y)). \end{aligned}$$

With $g(x) = \cos(cx)$ (so $g'(x) = -c\sin(cx)$), $c^2 g(x)g(y) \mp g'(x)g'(y) = c^2(\cos(cx)\cos(cy) \mp \sin(cx)\sin(cy)) = c^2\cos(c(x \mp y))$. Hence

$$\det H = c^4 \cos(c(x-y))\cos(c(x+y)),$$

which vanishes exactly on the lines $x - y = \text{const}$ and $x + y = \text{const}$ (specifically where $c(x \mp y) = \frac{\pi}{2} + k\pi$). These are two families of parallel lines, with slopes

$$\boxed{a = 1 \text{ and } a = -1}$$

(each family repeating with period $\pi\sqrt{2}/c$ along its perpendicular direction) — exactly the "more than one line" the problem describes, and this conclusion is forced by the request for the degenerate locus to be *straight lines*: for a product $g(x)g(y)$, $\det H$ factors into a function of $x - y$ times a function of $x + y$ only when $g'' = -c^2 g$.

(b) Yes. Besides the checkerboard of local maxima/minima at $g'(x) = g'(y) = 0$, f has genuine saddle points where $g(x) = g(y) = 0$ (e.g. $x = y = \pi/(2c)$ for $g = \cos(cx)$): there $f_{xx} = f_{yy} = 0$ but $f_{xy} = g'(x)g'(y) \neq 0$, so $\det H < 0$. On the diagonal $x = y$, $\nabla f = (-g(x)g'(x), -g(x)g'(x))$ always has equal components, so the diagonal is invariant under the gradient flow, and the induced one-dimensional flow $\dot{x} = g(x)g'(x)$ has the saddle at $x = \pi/(2c)$ as an *attracting* fixed point from the max at $x = 0$: any point (t, t) with $0 < t < \pi/(2c)$ has $\nabla f \neq 0$ (it vanishes only at $t = 0, \pi/(2c)$), yet gradient descent started there converges to the saddle, not to a minimum. So

$$\boxed{\text{yes, such points exist}}$$

— they lie on the stable manifold (here, a straight diagonal segment) of a saddle point. □

2.6 Vector Fields and Vector Calculus

Definitions and formulas.

- A vector field on $\mathbb{R}^n$ assigns a vector $F(x)$ to each point x. In coordinates, one writes

  $$F = (F_1, \dots, F_n).$$

  Vector fields model forces, velocities, gradients of scalar objectives, and fluxes.
- If ϕ is a scalar field, then its gradient

  $$\nabla\phi = \left(\frac{\partial\phi}{\partial x_1}, \dots, \frac{\partial\phi}{\partial x_n}\right)$$

  points in the direction of steepest increase. A field of the form $F = \nabla\phi$ is called *conservative*.
- For $F = (P, Q, R)$ in $\mathbb{R}^3$, the divergence and curl are

  $$\nabla \cdot F = \frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} + \frac{\partial R}{\partial z}, \qquad \nabla \times F = \left(\frac{\partial R}{\partial y} - \frac{\partial Q}{\partial z}, \frac{\partial P}{\partial z} - \frac{\partial R}{\partial x}, \frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right).$$

  Divergence measures local source or sink strength, while curl measures local rotation.
- The line integral of a vector field along a curve C is

  $$\int_C F \cdot dr.$$

  In mechanics this is the work done by a force field along the path.

- If $F = \nabla\phi$, then the fundamental theorem for line integrals gives

$$\int_C F \cdot dr = \phi(B) - \phi(A),$$

so the value depends only on the endpoints.

- Green's theorem is the planar bridge between line integrals and double integrals. If $C = \partial D$ is a positively oriented simple closed curve and $F = (P, Q)$ has continuously differentiable components, then

$$\oint_C P\,dx + Q\,dy = \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dA.$$

It converts circulation around the boundary into accumulated curl over the region. In flux form,

$$\oint_C F \cdot n\,ds = \iint_D \nabla \cdot F\,dA.$$

- The divergence theorem converts flux through a closed surface into a volume integral:

$$\iint_{\partial\Omega} F \cdot n\,dS = \iiint_\Omega \nabla \cdot F\,dV.$$

This is often the fastest route when the divergence is simple and the region has easy volume.

Example 2.9 (Flux of a radial field through the unit sphere)**.** Let

$$F(x, y, z) = (x, y, z).$$

Compute the outward flux of F through the unit sphere

$$x^2 + y^2 + z^2 = 1.$$

Solution. Use the divergence theorem with Ω equal to the unit ball. First compute

$$\nabla \cdot F = \frac{\partial x}{\partial x} + \frac{\partial y}{\partial y} + \frac{\partial z}{\partial z} = 3.$$

Therefore

$$\iint_{\partial\Omega} F \cdot n\,dS = \iiint_\Omega 3\,dV = 3\ \mathrm{Vol}(\Omega).$$

The unit ball has volume

$$\mathrm{Vol}(\Omega) = \frac{4\pi}{3},$$

so the flux is

$$3 \cdot \frac{4\pi}{3} = 4\pi.$$

Hence the outward flux is

$$\boxed{4\pi}.$$

As a quick check, on the unit sphere the outward unit normal is $n = (x, y, z)$, so $F \cdot n = x^2 + y^2 + z^2 = 1$, and the flux is just the surface area of the sphere, also equal to 4π. ☐

2.7 Integration, Parameter Integrals, and Special Functions

Definitions and formulas.

- Fundamental theorem of calculus:

$$\int_a^b f'(x)\,dx = f(b) - f(a).$$

- Integration by parts:

$$\int u\,dv = uv - \int v\,du.$$

- In a change of variables for multiple integrals,

$$\iint_D f(x,y)\,dx\,dy = \iint_{D'} f(x(u,v),y(u,v)) \left|\frac{\partial(x,y)}{\partial(u,v)}\right| du\,dv.$$

- Differentiation under the integral sign is often valid when a dominated convergence argument applies:

$$\frac{d}{d\theta}\int f(x,\theta)\,dx = \int \frac{\partial}{\partial\theta} f(x,\theta)\,dx.$$

- The Gamma and Beta functions are

$$\Gamma(s) = \int_0^\infty x^{s-1}e^{-x}\,dx, \qquad B(p,q) = \int_0^1 t^{p-1}(1-t)^{q-1}\,dt,$$

  with the identity

$$B(p,q) = \frac{\Gamma(p)\Gamma(q)}{\Gamma(p+q)}.$$

Example 2.10 (Gaussian integral and a normal fourth moment)**.** Evaluate

$$\int_{-\infty}^{\infty} e^{-x^2/2}\,dx$$

and compute $\mathbb{E}[X^4]$ for $X \sim \mathcal{N}(0,1)$.

Solution. Let

$$I = \int_{-\infty}^{\infty} e^{-x^2/2}\,dx.$$

Then

$$I^2 = \iint_{\mathbb{R}^2} e^{-(x^2+y^2)/2}\,dx\,dy.$$

In polar coordinates,

$$I^2 = \int_0^{2\pi}\int_0^\infty e^{-r^2/2} r\,dr\,d\theta = 2\pi \int_0^\infty e^{-r^2/2} r\,dr.$$

Set $u = r^2/2$, so $du = r\,dr$. Then

$$I^2 = 2\pi \int_0^\infty e^{-u}\,du = 2\pi.$$

Therefore

$$\boxed{I = \sqrt{2\pi}}.$$

Now the density of $X \sim \mathcal{N}(0,1)$ is

$$\phi(x) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2}.$$

So

$$\mathbb{E}[X^4] = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} x^4 e^{-x^2/2}\, dx.$$

Using integration by parts or the standard recursion for Gaussian moments,

$$\mathbb{E}[X^{2n}] = (2n-1)\mathbb{E}[X^{2n-2}], \qquad \mathbb{E}[X^0] = 1.$$

Hence

$$\mathbb{E}[X^4] = 3\mathbb{E}[X^2] = 3.$$

So

$$\boxed{\mathbb{E}[X^4] = 3}.$$

□

Example 2.11 (A nonlinear substitution with two branches)**.** Let $f : \mathbb{R} \to \mathbb{R}$ be integrable and suppose

$$\int_{\mathbb{R}} f(x)\, dx = 2026.$$

Find

$$\int_{\mathbb{R}} f\left(x - \frac{1}{x}\right) dx.$$

Solution. Set

$$g(x) = x - \frac{1}{x}, \qquad x \neq 0.$$

We split the real line into the two intervals $(-\infty, 0)$ and $(0, \infty)$. On each interval,

$$g'(x) = 1 + \frac{1}{x^2} > 0,$$

so g is strictly increasing. Moreover, each branch maps onto all of $\mathbb{R}$. Therefore every $y \in \mathbb{R}$ has exactly two preimages: one positive and one negative.

Let these inverse branches be $x_+(y) > 0$ and $x_-(y) < 0$, so

$$x_\pm(y) = \frac{y \pm \sqrt{y^2 + 4}}{2}.$$

They satisfy

$$x_+(y)x_-(y) = -1.$$

Now split the integral:

$$\int_{\mathbb{R}} f\left(x - \frac{1}{x}\right) dx = \int_0^{\infty} f(g(x))\, dx + \int_{-\infty}^{0} f(g(x))\, dx.$$

On each branch use the substitution $y = g(x)$. Since $dy = (1 + x^{-2})dx$, we get

$$dx = \frac{1}{1 + x^{-2}}\, dy.$$

Hence

$$\int_{\mathbb{R}} f\left(x - \frac{1}{x}\right) dx = \int_{\mathbb{R}} f(y) \frac{1}{1 + x_+(y)^{-2}}\, dy + \int_{\mathbb{R}} f(y) \frac{1}{1 + x_-(y)^{-2}}\, dy.$$

Using $x_-(y) = -1/x_+(y)$, we obtain

$$\frac{1}{1 + x_+(y)^{-2}} + \frac{1}{1 + x_-(y)^{-2}} = \frac{x_+(y)^2}{1 + x_+(y)^2} + \frac{1}{1 + x_+(y)^2} = 1.$$

Therefore

$$\int_{\mathbb{R}} f\left(x - \frac{1}{x}\right) dx = \int_{\mathbb{R}} f(y)\, dy = 2026.$$

So the answer is

$$\boxed{2026}.$$

□

Alternative solution via hyperbolic-sine substitution. Split the integral into positive and negative x:

$$\int_{\mathbb{R}} f\left(x - \frac{1}{x}\right) dx = \int_0^{\infty} f\left(x - \frac{1}{x}\right) dx + \int_{-\infty}^{0} f\left(x - \frac{1}{x}\right) dx.$$

On $(0, \infty)$ write $x = e^t$, so $dx = e^t dt$ and

$$x - \frac{1}{x} = e^t - e^{-t} = 2\sinh t.$$

On $(-\infty, 0)$ write $x = -e^{-t}$, so $dx = e^{-t} dt$ and again

$$x - \frac{1}{x} = -e^{-t} + e^t = 2\sinh t.$$

Therefore

$$\int_{\mathbb{R}} f\left(x - \frac{1}{x}\right) dx = \int_{\mathbb{R}} f(2\sinh t) e^t\, dt + \int_{\mathbb{R}} f(2\sinh t) e^{-t}\, dt = \int_{\mathbb{R}} f(2\sinh t)\, 2\cosh t\, dt.$$

Now substitute $u = 2\sinh t$, so $du = 2\cosh t\, dt$. This gives

$$\int_{\mathbb{R}} f\left(x - \frac{1}{x}\right) dx = \int_{\mathbb{R}} f(u)\, du = 2026.$$

This is the same identity as above, but the hyperbolic substitution packages both branches into one clean formula.

Example 2.12 (A vanishing-interval limit via rescaling)**.** Find

$$\lim_{x \to 0} \int_0^x \frac{\cos(t^3)}{t + x}\, dt.$$

Solution. Substitute $t = xu$, $dt = x\, du$; as t runs from 0 to x, u runs from 0 to 1 (for either sign of x), and $t + x = x(u+1)$, so

$$\int_0^x \frac{\cos(t^3)}{t + x}\, dt = \int_0^1 \frac{\cos(x^3 u^3)}{x(u+1)}\, x\, du = \int_0^1 \frac{\cos(x^3 u^3)}{u+1}\, du.$$

The integrand is bounded ($|\cos(x^3u^3)/(u+1)| \le 1$ on $[0,1]$) and, for each fixed $u \in [0,1]$, $\cos(x^3u^3) \to 1$ as $x \to 0$; by dominated convergence (or just uniform convergence on the compact interval $[0,1]$, valid here since $|\cos(x^3u^3) - 1| \le \frac{1}{2}x^6u^6 \to 0$ uniformly),

$$\lim_{x \to 0} \int_0^1 \frac{\cos(x^3u^3)}{u+1}\, du = \int_0^1 \frac{du}{u+1} = \ln 2.$$

So

$$\boxed{\lim_{x \to 0} \int_0^x \frac{\cos(t^3)}{t + x}\, dt = \ln 2}.$$

□

2.8 Series, Power Series, and Fourier Methods

Definitions and formulas.

- A necessary condition for convergence is

$$a_n \to 0.$$

  This is not sufficient: for example, $\sum 1/n$ diverges.

- Benchmark series to compare against:

$$\sum_{n=0}^{\infty} r^n \text{ converges if and only if } |r| < 1, \qquad \sum_{n=1}^{\infty} \frac{1}{n^p} \text{ converges if and only if } p > 1.$$

- Comparison test for nonnegative terms: if $0 \le a_n \le b_n$ for all large n and $\sum b_n$ converges, then $\sum a_n$ converges. If $0 \le b_n \le a_n$ for all large n and $\sum b_n$ diverges, then $\sum a_n$ diverges.

- Limit comparison test: if $a_n, b_n > 0$ and

$$\lim_{n\to\infty} \frac{a_n}{b_n} = c$$

  for some $c \in (0, \infty)$, then $\sum a_n$ and $\sum b_n$ have the same behavior.

- Integral test: if $a_n = f(n)$ where f is positive, continuous, and decreasing on $[1, \infty)$, then

$$\sum_{n=1}^{\infty} a_n \quad \text{and} \quad \int_1^{\infty} f(x)\,dx$$

  either both converge or both diverge.

- Ratio test:

$$L = \limsup_{n\to\infty} \left| \frac{a_{n+1}}{a_n} \right|.$$

  If $L < 1$, then $\sum a_n$ converges absolutely. If $L > 1$ or $L = \infty$, then $\sum a_n$ diverges. If $L = 1$, the test is inconclusive.

- Root test:

$$L = \limsup_{n\to\infty} |a_n|^{1/n}.$$

  If $L < 1$, then $\sum a_n$ converges absolutely. If $L > 1$, then $\sum a_n$ diverges. If $L = 1$, the test is inconclusive.

- Alternating-series test (Leibniz): if $b_n \downarrow 0$, then

$$\sum_{n=1}^{\infty} (-1)^{n-1} b_n$$

  converges.

- Absolute convergence means $\sum |a_n| < \infty$, and it always implies convergence of $\sum a_n$. A series may converge without being absolutely convergent; then it is called conditionally convergent.

- Dirichlet's test: if the partial sums $A_N = \sum_{n=1}^{N} a_n$ are uniformly bounded and b_n is monotone with $b_n \to 0$, then $\sum a_n b_n$ converges.

- Abel's test: if $\sum a_n$ converges and b_n is monotone and bounded, then $\sum a_n b_n$ converges.

- A power series
$$\sum_{n=0}^{\infty} c_n (x-a)^n$$
converges for $|x-a| < R$, where
$$R = \frac{1}{\limsup_{n\to\infty} |c_n|^{1/n}}.$$

- The geometric-series identities are used constantly:
$$\sum_{n=0}^{\infty} x^n = \frac{1}{1-x}, \qquad |x| < 1,$$
and differentiating gives
$$\sum_{n=1}^{\infty} n x^{n-1} = \frac{1}{(1-x)^2}.$$

- A Fourier series on $[-\pi, \pi]$ is
$$f(x) \sim \frac{a_0}{2} + \sum_{n\geq 1} (a_n \cos nx + b_n \sin nx),$$
with Parseval's identity
$$\frac{1}{\pi}\int_{-\pi}^{\pi} |f(x)|^2 dx = \frac{a_0^2}{2} + \sum_{n\geq 1} (a_n^2 + b_n^2).$$

Example 2.13 (Radius of convergence and sum)**.** Find the radius of convergence of
$$\sum_{n=1}^{\infty} n\left(\frac{x}{3}\right)^n,$$
and identify the sum for $|x| < 3$.

Solution. The coefficients are $c_n = n/3^n$. Since
$$\limsup_{n\to\infty} \left|\frac{n}{3^n}\right|^{1/n} = \frac{1}{3},$$
the radius of convergence is
$$R = 3.$$
Now set $y = x/3$. For $|y| < 1$,
$$\sum_{n=1}^{\infty} n y^n = y \sum_{n=1}^{\infty} n y^{n-1} = \frac{y}{(1-y)^2}.$$
Substitute back $y = x/3$ to obtain
$$\sum_{n=1}^{\infty} n\left(\frac{x}{3}\right)^n = \frac{x/3}{(1-x/3)^2} = \frac{3x}{(3-x)^2}, \qquad |x| < 3.$$
□

Example 2.14 (A recursively defined positive series)**.** Determine whether the series

$$\sum_{n=1}^{\infty} a_n$$

converges when

$$a_1 = 1, \qquad a_n = \sin(a_{n-1}) \quad \text{for } n \geq 2.$$

Solution. Since $a_1 = 1 > 0$ and $\sin x \in (0, x)$ for every $x \in (0, \pi)$, we get

$$0 < a_{n+1} = \sin(a_n) < a_n$$

for all n. Thus (a_n) is positive and decreasing, so it converges to some limit $L \geq 0$.

Passing to the limit in

$$a_{n+1} = \sin(a_n)$$

gives

$$L = \sin L.$$

The only nonnegative solution is $L = 0$, so $a_n \to 0$.

Now use the Taylor expansion near 0:

$$\sin x = x - \frac{x^3}{6} + O(x^5).$$

Hence

$$a_{n+1} = a_n - \frac{a_n^3}{6} + O(a_n^5).$$

Also,

$$\frac{a_{n+1}}{a_n} = \frac{\sin(a_n)}{a_n} \to 1.$$

Therefore

$$\frac{1}{a_{n+1}^2} - \frac{1}{a_n^2} = \frac{(a_n - a_{n+1})(a_n + a_{n+1})}{a_n^2 a_{n+1}^2} \to \frac{1}{3}.$$

So, after summing this asymptotic telescoping relation, we obtain

$$\frac{1}{a_n^2} \sim \frac{n}{3}.$$

Equivalently,

$$a_n \sim \sqrt{\frac{3}{n}}.$$

Thus the terms behave like a constant multiple of $n^{-1/2}$, and therefore

$$\sum_{n=1}^{\infty} a_n$$

has the same behavior as the divergent p-series

$$\sum_{n=1}^{\infty} \frac{1}{\sqrt{n}}.$$

Hence

$$\boxed{\sum_{n=1}^{\infty} a_n \text{ diverges.}}$$

□

3 Combinatorics

Combinatorics questions reward clean counting arguments, parity reasoning, and the ability to encode a counting problem algebraically.

3.1 Counting Rules, Inclusion–Exclusion, and Pigeonhole

Definitions and formulas.

- Product rule: if a task has a choices followed by b independent choices, there are ab outcomes.
- Sum rule: if two disjoint cases have a and b outcomes, there are $a+b$ outcomes.
- Pigeonhole principle: if more than n objects are placed into n boxes, some box contains at least two objects.
- Inclusion–exclusion for sets $A_1, \dots, A_n$:

$$\left|\bigcup_{i=1}^{n} A_i\right| = \sum |A_i| - \sum |A_i \cap A_j| + \sum |A_i \cap A_j \cap A_k| - \cdots .$$

- Stars and bars: the number of nonnegative integer solutions of

$$x_1 + \cdots + x_k = n$$

  is

$$\binom{n+k-1}{k-1}.$$

Example 3.1 (Stars and bars with upper bounds)**.** How many nonnegative integer solutions are there to

$$x_1 + x_2 + x_3 + x_4 = 10$$

subject to $x_i \le 4$ for every i?

Solution. Without the upper bounds, the count is

$$\binom{10+4-1}{4-1} = \binom{13}{3} = 286.$$

Now subtract solutions where some $x_i \ge 5$. For a fixed index, write $x_i = 5 + y_i$ with $y_i \ge 0$. Then the remaining sum is 5, giving

$$\binom{5+4-1}{3} = \binom{8}{3} = 56$$

solutions for each choice of i. There are 4 such choices.

If two variables are at least 5, say $x_i = 5 + y_i$ and $x_j = 5 + y_j$, then the remaining sum is 0, so there is exactly one solution. There are $\binom{4}{2} = 6$ such pairs. No three variables can each be at least 5 because the total would exceed 10.

Therefore, by inclusion–exclusion,

$$286 - 4 \cdot 56 + 6 \cdot 1 = 286 - 224 + 6 = 68.$$

So the answer is

$$\boxed{68}.$$

□

Example 3.2 (Counting surjections by inclusion–exclusion)**.** How many surjective functions are there from a 5-element set onto a 3-element set?

Solution. Let the codomain be $\{1, 2, 3\}$. There are

$$3^5$$

functions in total.

Now subtract the functions that miss at least one codomain value. If a function misses a particular value, then all 5 elements map into the remaining 2 values, so there are

$$2^5$$

such functions. Since there are 3 choices for the missing value, this contributes $3 \cdot 2^5$.

But functions that use only one value were subtracted twice, so we add them back. There are 3 constant functions.

Therefore, by inclusion–exclusion, the number of surjections is

$$3^5 - 3 \cdot 2^5 + 3 = 243 - 96 + 3 = 150.$$

Hence

$$\boxed{150}.$$

□

Example 3.3 (Does $n + \tau(n)$ hit almost every integer?)**.** Let $\tau(n)$ denote the number of divisors of n. Is it true that all but finitely many positive integers can be written as $n + \tau(n)$ for some n?

Solution. No. Write $f(n) = n+\tau(n)$; since $\tau(n) \geq 1$, $f(n) \geq n+1$, so $m = 1$ is never representable — a first, trivial exception. The real obstruction is that f is far from injective, because τ itself is extremely erratic: it equals 2 at every prime but jumps to $3, 4, 5, \ldots$ at nearby composites, so nearby inputs $n_1 < n_2$ routinely satisfy $n_2 - n_1 = \tau(n_1) - \tau(n_2)$, i.e. $f(n_1) = f(n_2)$. For instance,

$$f(4) = f(5) = 7, \qquad f(8) = f(9) = 12, \qquad f(12) = f(14) = 18, \qquad f(30) = f(32) = f(34) = 38.$$

Every such collision consumes two (or more) inputs n to produce only one output value m, and since the domain $\{1, \ldots, N\}$ has only N elements while f must (for large N) cover an interval of about N target integers, persistent collisions of this kind force the image to miss a comparable number of targets rather than merely finitely many. Direct computation confirms this is not a boundary effect: of the integers up to 2,000,000, about 33% (roughly 660,000) are *not* of the form $n + \tau(n)$, and the proportion does not shrink in later windows (e.g. it is about 33% again in $(1{,}950{,}000, 2{,}000{,}000]$) — consistent with a positive density of exceptions, far more than "finitely many." So

$$\boxed{\text{no}}$$

— infinitely many positive integers (in fact a positive proportion of them) are not of the form $n + \tau(n)$. □

3.2 Permutations, Combinations, and Hypergeometric Counting

Definitions and formulas.

- Ordered selections without repetition:

$$A_n^k = \frac{n!}{(n-k)!}.$$

- Unordered selections without repetition:
$$\binom{n}{k} = \frac{n!}{k!(n-k)!}.$$
- Permutations with repeated letters are counted by multinomial coefficients:
$$\frac{n!}{n_1!\cdots n_r!}.$$
- Newton's binomial theorem:
$$(x+y)^n = \sum_{k=0}^{n} \binom{n}{k} x^k y^{n-k}.$$
- In the hypergeometric model, if a population has N items, K of them are marked, and we draw n items without replacement, then
$$\mathbb{P}(X=r) = \frac{\binom{K}{r}\binom{N-K}{n-r}}{\binom{N}{n}}.$$

Example 3.4 (Exactly two aces)**.** What is the probability that a 5-card poker hand contains exactly two aces?

Solution. There are $\binom{52}{5}$ equally likely hands. To get exactly two aces, choose 2 of the 4 aces and 3 of the 48 non-aces:
$$\mathbb{P}(\text{exactly two aces}) = \frac{\binom{4}{2}\binom{48}{3}}{\binom{52}{5}}.$$
That is the final answer. Numerically it is approximately 0.03993. □

Example 3.5 (Expected number of fixed points)**.** If π is a uniformly random permutation of $\{1, 2, \dots, n\}$, what is the expected number of indices i such that $\pi(i) = i$?

Solution. Let
$$X = \#\{i : \pi(i) = i\}$$
be the number of fixed points. For each i, define the indicator
$$I_i = \begin{cases} 1, & \text{if } \pi(i) = i, \\ 0, & \text{otherwise.} \end{cases}$$
Then
$$X = I_1 + \cdots + I_n.$$
For a uniformly random permutation, every value is equally likely to appear in position i, so
$$\mathbb{P}(\pi(i) = i) = \frac{1}{n}.$$
Hence
$$\mathbb{E}[I_i] = \frac{1}{n}.$$
By linearity of expectation,
$$\mathbb{E}[X] = \sum_{i=1}^{n} \mathbb{E}[I_i] = n \cdot \frac{1}{n} = 1.$$
Therefore the expected number of fixed points is
$$\boxed{1}.$$
□

3.3 Graphs, Trees, Planarity, and Tournaments

Definitions and formulas.

- Handshake lemma:
$$\sum_{v\in V}\deg(v) = 2|E|.$$
In particular, the number of odd-degree vertices is even.
- A tree on n vertices has exactly $n-1$ edges. A connected graph with $n-1$ edges is a tree.
- A graph has an Eulerian circuit if and only if every vertex has even degree and the non-isolated part is connected.
- Euler's formula for connected planar graphs is
$$V - E + F = 2.$$
- Every tournament has a Hamiltonian path.

Example 3.6 (Parity-constrained graphs on 100 vertices)**.** How many simple graphs on 100 labeled vertices have an even number of edges and all degrees odd?

Solution. Let $m = \binom{100}{2} = 4950$ be the number of possible edges, and identify a graph with its edge-indicator vector $x \in \mathbb{F}_2^m$.

Define the incidence map over $\mathbb{F}_2$,
$$M : \mathbb{F}_2^m \to \mathbb{F}_2^{100},$$
where Mx records the parity of the degrees. For the complete graph K_{100}, the incidence matrix has rank 99 over $\mathbb{F}_2$, because the graph is connected and the image is exactly the set of parity vectors with even total sum.

The requirement "all degrees odd" means
$$Mx = (1, 1, \ldots, 1).$$
This right-hand side is feasible because the sum of its coordinates is $100 \equiv 0 \pmod 2$. Therefore the set of solutions is an affine space of dimension
$$4950 - 99 = 4851.$$
So there are 2^{4851} graphs with all degrees odd.

Now impose the additional condition that the total number of edges is even. This is one more linear equation over $\mathbb{F}_2$:
$$\sum_e x_e = 0.$$
This equation is independent of the degree-parity equations: if it were a linear combination of them, there would exist bits $y_1, \ldots, y_{100}$ such that $y_i + y_j = 1$ for every pair $i \neq j$, which is impossible for $100 \geq 3$.

Therefore the extra parity condition cuts the affine space in half. The final count is
$$\boxed{2^{4850}}.$$
□

Example 3.7 (More than n^2 edges on $2n$ vertices force a diamond)**.** Show that every graph on $2n$ vertices with $n^2 + 1$ edges contains a diamond, that is, a copy of K_4 with one edge removed.

Solution. We prove the stronger statement that every diamond-free graph on N vertices has at most

$$\left\lfloor \frac{N^2}{4} \right\rfloor$$

edges. Taking $N = 2n$ will then give the desired conclusion.

Proceed by induction on N. The claim is trivial for $N \leq 3$.

Now let G be a diamond-free graph on $N \geq 4$ vertices.

If G is triangle-free, then Mantel's theorem applies directly and gives

$$e(G) \leq \left\lfloor \frac{N^2}{4} \right\rfloor.$$

Mantel's theorem used here. Mantel's theorem states that every triangle-free graph on N vertices has at most

$$\left\lfloor \frac{N^2}{4} \right\rfloor$$

edges, with equality for the complete bipartite graph $K_{\lfloor N/2 \rfloor, \lceil N/2 \rceil}$. One short proof is this: if uv is an edge in a triangle-free graph, then u and v have no common neighbor, so

$$\deg(u) + \deg(v) \leq N.$$

Summing over all edges gives

$$\sum_v \deg(v)^2 = \sum_{uv \in E} (\deg(u) + \deg(v)) \leq N e(G).$$

Now apply Cauchy to the degree sequence:

$$(2e(G))^2 = \left(\sum_v \deg(v) \right)^2 \leq N \sum_v \deg(v)^2 \leq N^2 e(G).$$

Cancelling $e(G)$ yields $e(G) \leq N^2/4$, hence $e(G) \leq \lfloor N^2/4 \rfloor$.

So the only remaining case is that G contains a triangle T. Write $T = \{a, b, c\}$. Since G is diamond-free, no vertex outside T can be adjacent to two vertices of T: if some outside vertex v were adjacent to, say, a and b, then the four vertices a, b, c, v would contain two triangles sharing the edge ab, which is exactly a diamond.

Therefore every vertex outside T sends at most one edge into T. As there are $N - 3$ outside vertices, the number of edges joining T to $V(G) \setminus T$ is at most $N - 3$. Adding the 3 edges of the triangle itself, we find that the total number of edges incident with T is at most

$$(N - 3) + 3 = N.$$

Hence

$$e(G) \leq e(G - T) + N.$$

Now $G - T$ has $N - 3$ vertices and is still diamond-free, so by the induction hypothesis,

$$e(G - T) \leq \left\lfloor \frac{(N-3)^2}{4} \right\rfloor.$$

Thus

$$e(G) \leq \left\lfloor \frac{(N-3)^2}{4} \right\rfloor + N.$$

It remains to compare this with $\lfloor N^2/4 \rfloor$. If $N = 2m$, then

$$\left\lfloor \frac{(N-3)^2}{4} \right\rfloor + N = \left\lfloor \frac{(2m-3)^2}{4} \right\rfloor + 2m = m^2 - m + 2 \le m^2 = \left\lfloor \frac{N^2}{4} \right\rfloor$$

for $m \ge 2$. If $N = 2m+1$, then

$$\left\lfloor \frac{(N-3)^2}{4} \right\rfloor + N = \frac{(2m-2)^2}{4} + (2m+1) = m^2 + 2 \le m^2 + m = \left\lfloor \frac{N^2}{4} \right\rfloor$$

for $m \ge 2$. So in all cases,

$$e(G) \le \left\lfloor \frac{N^2}{4} \right\rfloor.$$

Now specialize to $N = 2n$. A diamond-free graph on $2n$ vertices has at most

$$\left\lfloor \frac{(2n)^2}{4} \right\rfloor = n^2$$

edges. Therefore any graph on $2n$ vertices with $n^2 + 1$ edges cannot be diamond-free. Hence it must contain a diamond.

So the answer is

$$\boxed{\text{every graph on } 2n \text{ vertices with } n^2+1 \text{ edges contains a diamond.}}$$

□

Example 3.8 (Hamiltonian path in a tournament)**.** In a tournament on n vertices, prove that there exists a directed path that visits every vertex exactly once.

Solution. Use induction on n. The claim is obvious for $n = 1$.

Assume it holds for tournaments on $n-1$ vertices, and consider a tournament on vertices $v_1, \dots, v_n$. Remove one vertex, say v_n. By the induction hypothesis, the remaining tournament has a Hamiltonian path

$$u_1 \to u_2 \to \cdots \to u_{n-1}.$$

Now insert v_n into this path. If $v_n \to u_1$, place v_n at the beginning. If $u_{n-1} \to v_n$, place v_n at the end. Otherwise there exists an index k such that

$$u_k \to v_n \to u_{k+1}.$$

Indeed, scan the path from left to right until the direction of the edge involving v_n changes. Inserting v_n between u_k and u_{k+1} produces a Hamiltonian path through all n vertices.

Thus every tournament has a Hamiltonian path. □

Example 3.9 (Roads and cities as a finite incidence structure)**.** Suppose there are n roads and finitely many cities such that

- any two distinct cities lie on a road;
- any two distinct roads meet in exactly one city.

How many cities can there be?

Solution. Let the number of cities be m.

First note that any two cities actually lie on a *unique* road. Indeed, if two different roads both passed through the same two cities, then those two roads would meet in at least two cities, contradicting the assumption that any two roads meet in exactly one city.

So we have a finite incidence structure in which any two cities determine a unique road. A classical theorem of de Bruijn and Erdős says that in any finite linear space, the number of lines is at least the number of points. Applying it here gives

$$n \ge m.$$

Sketch of the de Bruijn–Erdős theorem used here. Write P for the set of points and L for the set of lines in a finite linear space, and let

$$v = |P|, \qquad b = |L|.$$

For a point p, let $d(p)$ be the number of lines through p, and for a line ℓ, let $s(\ell)$ be the number of points on ℓ.

The key inequality is that if $p \notin \ell$, then

$$s(\ell) \leq d(p),$$

because each point of ℓ together with p determines a distinct line through p.

Now assume $b \leq v$. A standard Hall-marriage argument, applied to the family of complements $P \setminus \ell$, gives distinct points $p_\ell \notin \ell$ for each line $\ell \in L$. Therefore

$$s(\ell) \leq d(p_\ell) \qquad \text{for every } \ell \in L.$$

Hall's theorem used here. Hall's marriage theorem says that a finite family of sets $S_1, \dots, S_m$ has a system of distinct representatives if and only if for every index set $I \subseteq \{1, \dots, m\}$,

$$\left| \bigcup_{i \in I} S_i \right| \geq |I|.$$

Here we apply it to the sets $S_\ell = P \setminus \ell$. Why does Hall's condition hold? In a nontrivial linear space, every line misses at least one point, so the case $|I| = 1$ is clear. If $2 \leq |I| < b$, then the corresponding distinct lines have common intersection of size at most 1, hence

$$\left| \bigcup_{\ell \in I} (P \setminus \ell) \right| = \left| P \setminus \bigcap_{\ell \in I} \ell \right| \geq v - 1 \geq b - 1 \geq |I|.$$

If $|I| = b$, then the union is all of P, so it has size $v \geq b$. Thus Hall's theorem produces distinct points $p_\ell \notin \ell$ as claimed.

Summing over all lines yields

$$\sum_{\ell \in L} s(\ell) \leq \sum_{\ell \in L} d(p_\ell) \leq \sum_{p \in P} d(p).$$

But both outer sums count the same thing, namely the total number of incidences point–line, so

$$\sum_{\ell \in L} s(\ell) = \sum_{p \in P} d(p).$$

Hence every inequality above is in fact an equality, which forces $b = v$ whenever $b \leq v$. Therefore in all cases

$$b \geq v.$$

That is the de Bruijn–Erdős theorem in the form used here.

Now use duality: interchange the roles of cities and roads. The second assumption says that any two roads meet in exactly one city, so in the dual picture any two "points" determine a unique "line". Applying the same theorem to the dual incidence structure gives

$$m \geq n.$$

Combining the two inequalities yields

$$\boxed{m = n}.$$

So the number of cities must be exactly the number of roads. □

Example 3.10 (Expected Number of Cliques)**. Problem:** There are n students seeing each other for the first time. An amusement park offers a discount if a group of exactly k friends comes. Any two random students can become friends with probability p. On average, how many such valid groups of exactly k friends can be formed?

Solution: Let the n students be the vertices of a random graph $G(n, p)$, where each edge exists independently with probability p. A valid group of exactly k friends corresponds to a k-clique (a complete subgraph K_k) in the graph.

Let S be a specific subset of k students. We define an indicator random variable I_S such that $I_S = 1$ if S forms a k-clique, and 0 otherwise. A subset of size k contains exactly $\binom{k}{2}$ possible pairs. For all these pairs to be friends, every corresponding edge must be present. Thus,

$$P(I_S = 1) = p^{\binom{k}{2}}, \quad \text{and hence} \quad E[I_S] = p^{\binom{k}{2}}.$$

The total number of k-cliques X is the sum of I_S over all possible subsets S of size k. By linearity of expectation,

$$E[X] = E\left[\sum_{S\subset V, |S|=k} I_S\right] = \sum_{|S|=k} E[I_S] = \binom{n}{k} p^{\binom{k}{2}}.$$

Example 3.11 (Guaranteed Dominating Set via Maximal Independent Sets)**. Problem:** There are 100 people in a company. It is known that among any 10 people, there are three pairwise acquaintances. What is the smallest n that guarantees there are n people such that any of the remaining ones is acquainted with someone in these n?

Solution: We model the people as the vertices V of a graph G, where edges represent acquaintances. The condition states that any subset of 10 vertices contains a triangle (a K_3).

We are looking for the smallest n that guarantees the existence of a dominating set of size at most n. (A set $D \subseteq V$ is dominating if every vertex in $V \setminus D$ has at least one neighbor in D).

We first claim that the independence number satisfies $\alpha(G) \leq 8$. Indeed, suppose there were 9 pairwise nonadjacent vertices $v_1, \dots, v_9$. Take any tenth vertex w. Then the induced subgraph on

$$\{v_1, \dots, v_9, w\}$$

contains no triangle: a triangle cannot lie entirely inside $\{v_1, \dots, v_9\}$ because that set is independent, and any triangle using w would require an edge between two of the vertices v_i, which also does not exist. This contradicts the hypothesis that every 10-vertex subset contains a triangle. Hence

$$\alpha(G) \leq 8.$$

Now, consider any *maximal* independent set I in G. By maximality, we cannot add any other vertex from $V \setminus I$ to I without creating an edge. This implies that every vertex in $V \setminus I$ must be adjacent to at least one vertex in I. Consequently, every maximal independent set is also a dominating set.

Since every independent set has size at most 8, a maximal independent set I has size $|I| \leq 8$. So G always has a dominating set of size at most 8.

This bound is sharp. Consider a graph consisting of one clique on 93 vertices together with 7 isolated vertices. Any set of 10 vertices must contain at least 3 vertices from the 93-clique, and those 3 vertices form a triangle. So the hypothesis is satisfied. On the other hand, every dominating set must contain all 7 isolated vertices, and it must also contain at least one vertex from the clique in order to dominate the clique vertices. Therefore every dominating set has size at least

$$7 + 1 = 8.$$

Hence the smallest value of n that is guaranteed is

$$\boxed{8}.$$

Example 3.12 (Minimum edges forced by a local domination condition)**.** A graph G has 40 vertices, and among any 5 of them there is always one adjacent to the other 4. What is the minimum possible number of edges of G?

Solution. Pass to the complement $H = \overline{G}$; since $|E(G)| + |E(H)| = \binom{40}{2} = 780$, minimizing $|E(G)|$ is the same as maximizing $|E(H)|$. A vertex v is G-adjacent to the other four vertices of a 5-set S exactly when v has no H-neighbor inside S, i.e. v is isolated in the induced subgraph $H[S]$. So the hypothesis on G says: every 5-element $S \subseteq V(H)$ contains a vertex isolated in $H[S]$. We must maximize $|E(H)|$ subject to this.

No component of H has 5 *or more vertices.* If a component C had $|C| \geq 5$, a spanning tree of C has a connected sub-tree on some 5 vertices S (e.g. the first 5 vertices visited by a breadth-first search). The tree edges among S are genuine edges of H, so every vertex of S has an H-neighbor inside S: no isolated vertex, contradicting the hypothesis.

H cannot have two distinct components of sizes ≥ 2 *and* ≥ 3. If C_1, C_2 are distinct components with $|C_1| \geq 2$, $|C_2| \geq 3$, take an edge $\{a, b\} \subseteq C_1$ and three vertices $\{c, d, e\} \subseteq C_2$ that induce a subgraph with no isolated point (any connected graph on ≥ 3 vertices contains such a triple, e.g. a vertex together with two of its tree-neighbors). Then $S = \{a, b, c, d, e\}$ has no isolated vertex in $H[S]$, a contradiction.

So the components of H fall into one of two patterns: (A) a single component of size $s \in \{2, 3, 4\}$ plus $40-s$ isolated vertices, contributing at most $\binom{4}{2} = 6$ edges; or (B) every component has size at most 2 (a disjoint union of edges and isolated vertices), contributing at most $\lfloor 40/2 \rfloor = 20$ edges. Since $20 > 6$, the maximum is achieved by a perfect matching on all 40 vertices, $|E(H)| = 20$.

Thus

$$|E(G)| = 780 - 20 = \boxed{760},$$

attained by $G = \overline{H} = K_{40}$ minus a perfect matching (each vertex has exactly one non-neighbor, its "partner"). Indeed, in any 5 chosen vertices, since 5 is odd the partners cannot all pair up inside the set, so some vertex's partner lies outside it — that vertex is adjacent to the other four. □

Example 3.13 (The double-sweep diameter heuristic can fail forever)**.** For a connected graph, the following classical heuristic estimates the diameter: set A_0 = a fixed start vertex; run BFS from A_0 and let A_1 be a vertex at maximum distance from A_0; run BFS from A_1 and let A_2 be a vertex at maximum distance from A_1 (breaking all ties by smallest vertex label); continue to build $A_3, A_4, \ldots$, and estimate the diameter as $\max_i \operatorname{dist}(A_i, A_{i+1})$. Exhibit a connected graph on at most 10 vertices, with A_0 a fixed labeled vertex, on which this process — however long it is run — never produces the true diameter.

Solution. This heuristic ("double sweep") is exact on trees, so any counterexample must contain a cycle. Take the 5-vertex graph consisting of the 4-cycle $1-2-3-4-1$ with a pendant vertex 5 attached to vertex 2 (edges $\{1,2\}, \{2,3\}, \{3,4\}, \{1,4\}, \{2,5\}$), and start the process at $A_0 = 1$. Direct computation gives the distance matrix

	1	2	3	4	5
1	0	1	2	1	2
2	1	0	1	2	1
3	2	1	0	1	2
4	1	2	1	0	3
5	2	1	2	3	0

so the true diameter is 3, uniquely realized by the pair $(4, 5)$.

Now run the process. From $A_0 = 1$, the farthest vertices are 3 and 5, both at distance 2; the tie-break picks the smaller label, so $A_1 = 3$. From $A_1 = 3$, the farthest vertices are 1 and 5, again both at distance 2, and the tie-break sends us back to $A_2 = 1$. The process has entered the cycle $1 \to 3 \to 1 \to 3 \to \cdots$ and will repeat it forever, with $\mathrm{dist}(1,3) = \mathrm{dist}(3,1) = 2$ at every step. Hence

$$\max_i \mathrm{dist}(A_i, A_{i+1}) = 2 < 3 = \text{true diameter},$$

for every stopping point k — the diametral pair $(4,5)$ is never even visited as a sweep center, so

$$\boxed{\text{no choice of } k \text{ recovers the diameter on this graph}}.$$

□

3.4 Generating Functions and Linear Recurrences

Definitions and formulas.

- The ordinary generating function of a sequence (a_n) is

$$G(x) = \sum_{n=0}^{\infty} a_n x^n.$$

- Shifts in the sequence translate into simple identities. For example,

$$\sum_{n=0}^{\infty} a_{n+1} x^n = \frac{G(x) - a_0}{x}, \qquad \sum_{n=0}^{\infty} a_{n+2} x^n = \frac{G(x) - a_0 - a_1 x}{x^2}.$$

- A linear recurrence of order d with constant coefficients has the form

$$a_{n+d} = c_1 a_{n+d-1} + \cdots + c_d a_n.$$

  The associated characteristic equation is

$$r^d - c_1 r^{d-1} - \cdots - c_d = 0.$$

- If the characteristic roots $r_1, \ldots, r_m$ are distinct, then the homogeneous solution has the form

$$a_n = \alpha_1 r_1^n + \cdots + \alpha_m r_m^n.$$

  If a root r has multiplicity k, then its contribution becomes

$$(\beta_0 + \beta_1 n + \cdots + \beta_{k-1} n^{k-1}) r^n.$$

- For nonhomogeneous recurrences, look for a particular solution matching the forcing term, then add the homogeneous solution.
- For constant-coefficient recurrences, the generating function is rational; the denominator encodes the recurrence relation.

Example 3.14 (Characteristic-root method for a linear recurrence)**.** Solve the recurrence

$$a_{n+2} = 5a_{n+1} - 6a_n, \qquad a_0 = 2, \qquad a_1 = 5.$$

Solution. The characteristic equation is

$$r^2 - 5r + 6 = 0,$$

which factors as

$$(r-2)(r-3)=0.$$

Hence the general solution is

$$a_n = \alpha 2^n + \beta 3^n.$$

Use the initial conditions:

$$a_0 = \alpha + \beta = 2, \qquad a_1 = 2\alpha + 3\beta = 5.$$

Subtracting twice the first equation from the second gives $\beta = 1$, so $\alpha = 1$. Therefore

$$\boxed{a_n = 2^n + 3^n}.$$

This is the standard template for linear recurrences with distinct characteristic roots. □

Example 3.15 (Fibonacci generating function)**.** Let $F_0 = 0$, $F_1 = 1$, and $F_{n+2} = F_{n+1} + F_n$. Find the generating function

$$G(x) = \sum_{n=0}^{\infty} F_n x^n.$$

Solution. Multiply the recurrence by x^{n+2} and sum over $n \geq 0$:

$$\sum_{n\geq 0} F_{n+2}x^{n+2} = \sum_{n\geq 0} F_{n+1}x^{n+2} + \sum_{n\geq 0} F_n x^{n+2}.$$

The left side is $G(x) - x$. The first right-side term is $xG(x)$, and the second is $x^2G(x)$. Therefore

$$G(x) - x = xG(x) + x^2G(x).$$

Rearranging gives

$$G(x)(1 - x - x^2) = x,$$

so

$$\boxed{G(x) = \frac{x}{1 - x - x^2}}.$$

□

4 Probability Theory and Statistics

This is the most frequently tested area in quant interviews. You are expected to move comfortably between combinatorics, conditional probability, expectation, convergence, and recursion.

4.1 Probability Spaces, Conditioning, and Bernoulli Schemes

Definitions and formulas.

- A probability space is $(\Omega, \mathcal{F}, \mathbb{P})$, where Ω is the sample space, $\mathcal{F}$ is a sigma-algebra, and $\mathbb{P}$ is a probability measure.
- Conditional probability:
$$\mathbb{P}(A \mid B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)}.$$
- Total probability and Bayes' rule:
$$\mathbb{P}(B) = \sum_i \mathbb{P}(B \mid A_i)\mathbb{P}(A_i), \qquad \mathbb{P}(A_k \mid B) = \frac{\mathbb{P}(B \mid A_k)\mathbb{P}(A_k)}{\sum_i \mathbb{P}(B \mid A_i)\mathbb{P}(A_i)}.$$

- In a Bernoulli scheme, if $X \sim \mathrm{Bin}(n,p)$ then

$$\mathbb{P}(X=k) = \binom{n}{k} p^k (1-p)^{n-k}, \qquad \mathbb{E}[X] = np, \qquad \mathrm{Var}(X) = np(1-p).$$

- If $n \to \infty$ and $p \to 0$ with $np \to \lambda$, then the binomial law converges to the Poisson law:

$$\mathbb{P}(X=k) \to e^{-\lambda} \frac{\lambda^k}{k!}.$$

Example 4.1 (Bayes' rule in a rare-event setting)**.** A test for a disease has sensitivity 95% and specificity 90%. The disease prevalence is 1%. If a randomly chosen person tests positive, what is the probability that the person actually has the disease?

Solution. Let D be the event "has the disease" and $+$ be the event "tests positive". Then

$$\mathbb{P}(D) = 0.01, \qquad \mathbb{P}(+ \mid D) = 0.95, \qquad \mathbb{P}(+ \mid D^c) = 0.10.$$

By Bayes' rule,

$$\mathbb{P}(D \mid +) = \frac{0.95 \cdot 0.01}{0.95 \cdot 0.01 + 0.10 \cdot 0.99} = \frac{0.0095}{0.1085} \approx 0.0876.$$

So the posterior probability is about

$$\boxed{8.76\%}.$$

This is a standard reminder that in low-prevalence settings, false positives can dominate. □

Example 4.2 (Meeting probability and a non-obvious dependence)**.** Alex and Maria agree to meet between 8:00 and 9:00. Each arrives at a uniformly random, independent moment in that hour, waits 15 minutes, and leaves. Let A be the event that they do not meet, and B the event that at least one of them arrives after 8:45. Are A and B independent?

Solution. Let $X, Y \sim \mathrm{Unif}(0, 60)$ (minutes past 8:00) be independent arrival times. They meet iff their 15-minute waiting windows overlap, i.e. $|X - Y| \leq 15$, so

$$A = \{|X-Y| > 15\}, \qquad B = \{X > 45\} \cup \{Y > 45\}.$$

Rescale to the unit square, $u = X/60$, $v = Y/60$: $A = \{|u-v| > 1/4\}$, $B = \{u > 3/4\} \cup \{v > 3/4\}$.

The region A is two congruent corner triangles, each of area $\frac{1}{2}(\frac{3}{4})^2 = \frac{9}{32}$, so $\mathbb{P}(A) = \frac{9}{16}$ and $\mathbb{P}(B) = 1 - (\frac{3}{4})^2 = \frac{7}{16}$.

Consider $T = \{v > u + 1/4\}$ (the other triangle, $\{u > v + 1/4\}$, is its mirror image). Inside T, u never exceeds $3/4$, so $B \cap T = \{v > 3/4\} \cap T$. Splitting at $u = 1/2$ (where $u + 1/4$ crosses $3/4$),

$$\mathrm{area}(T \cap B) = \int_0^{1/2} \frac{1}{4}\, du + \int_{1/2}^{3/4} \left(\frac{3}{4} - u\right) du = \frac{1}{8} + \frac{1}{32} = \frac{5}{32}.$$

By the symmetry $u \leftrightarrow v$ (which fixes B), the mirror triangle contributes the same area, so

$$\mathbb{P}(A \cap B) = 2 \cdot \frac{5}{32} = \frac{5}{16} = \frac{80}{256}.$$

Comparing with $\mathbb{P}(A)\mathbb{P}(B) = \frac{9}{16} \cdot \frac{7}{16} = \frac{63}{256}$,

$$\boxed{A \text{ and } B \text{ are } \textit{not} \text{ independent}}$$

(not meeting is positively correlated with a late arrival, which matches intuition: a late arrival shrinks the overlap of the two waiting windows). □

Example 4.3 (A two-stage secret-gift-exchange probability)**.** A club of $3n$ people prepares gifts for a party: exactly n people want a tie, n want socks, and n want a toy dinosaur. Each person buys, uniformly at random and independently, one of the two gift types they do *not* want themselves. All $3n$ gifts are pooled and then redistributed to the $3n$ people via a uniformly random bijection. Alice (who wants a dinosaur) and Bob (who wants socks) are both in the club. Find the probability that neither Alice nor Bob receives the gift type they wanted.

Solution. **Stage 1: the pool composition.** Let a = number of tie-wanters who buy socks (so $n-a$ buy a dinosaur), b = number of socks-wanters who buy a dinosaur (so $n-b$ buy a tie), c = number of dinosaur-wanters who buy a tie (so $n-c$ buy socks); a, b, c are i.i.d. Binomial$(n, \frac{1}{2})$. The pool then contains

$$s = a + (n-c) \text{ socks}, \qquad d = (n-a) + b \text{ dinosaurs},$$

(and $3n - s - d$ ties), with $\mathbb{E}[a] = \mathbb{E}[b] = \mathbb{E}[c] = n/2$, so $\mathbb{E}[s] = \mathbb{E}[d] = n$.

Stage 2: the redistribution. Given the pool composition (s, d), drawing Alice's and Bob's gifts is equivalent to sampling 2 gifts without replacement from the $3n$ pooled gifts. By inclusion–exclusion,

$$\mathbb{P}(\text{Alice} \neq d,\ \text{Bob} \neq s \mid s, d) = 1 - \frac{d}{3n} - \frac{s}{3n} + \frac{ds}{3n(3n-1)}.$$

Combining. Taking expectation over the pool (law of total probability), we need $\mathbb{E}[d]$, $\mathbb{E}[s]$ (already found) and $\mathbb{E}[ds]$. Writing $ds = [(n-a)+b][a+(n-c)]$ and expanding using independence of a, b, c:

$$\mathbb{E}[ds] = \mathbb{E}[(n-a)a] + \mathbb{E}[(n-a)(n-c)] + \mathbb{E}[ab] + \mathbb{E}[b(n-c)] = \left(\frac{n^2}{4} - \frac{n}{4}\right) + \frac{n^2}{4} + \frac{n^2}{4} + \frac{n^2}{4} = n^2 - \frac{n}{4},$$

using $\mathbb{E}[(n-a)a] = n\mathbb{E}[a] - \mathbb{E}[a^2] = \frac{n^2}{2} - (\operatorname{Var}(a) + \mathbb{E}[a]^2) = \frac{n^2}{2} - (\frac{n}{4} + \frac{n^2}{4}) = \frac{n^2}{4} - \frac{n}{4}$, and the other three terms factor by independence.

So

$$\mathbb{P}(\text{neither gets what they wanted}) = 1 - \frac{n}{3n} - \frac{n}{3n} + \frac{n^2 - n/4}{3n(3n-1)} = \frac{1}{3} + \frac{4n-1}{12(3n-1)},$$

that is,

$$\boxed{\frac{16n-5}{12(3n-1)}}.$$

(For $n = 1$ this gives $11/24$, which a direct enumeration over the 3-person case confirms exactly.) $\square$

4.2 Random Variables, Moments, and Common Distributions

Definitions and formulas.

- A random variable X is a measurable function from Ω to $\mathbb{R}$. Its cumulative distribution function is
$$F_X(x) = \mathbb{P}(X \le x).$$
If a density exists, then $F'_X(x) = f_X(x)$ almost everywhere.
- The *distribution* of X is the full probabilistic description of X. It tells you which values are typical, how much uncertainty there is around the mean, whether the support is bounded or unbounded, and how quickly tail probabilities decay.

- In interview language, saying "X has distribution $\mathcal{D}$" should answer two questions at once: what real-world mechanism X models, and which structural property of the law matters most (counting, waiting time, symmetry, bounded support, tail behavior, or conjugacy).
- Expectation, variance, and covariance are

$$\mathbb{E}[X], \qquad \mathrm{Var}(X) = \mathbb{E}[X^2] - \mathbb{E}[X]^2, \qquad \mathrm{Cov}(X, Y) = \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y].$$

- Moments summarize a distribution, but the distribution contains more information than finitely many moments. In practice, quant questions often hinge on support, symmetry, memorylessness, or the tail rather than just the mean and variance.
- If $Y = AX + b$ for a random vector X, then

$$\mathbb{E}[Y] = A\mathbb{E}[X] + b, \qquad \mathrm{Cov}(Y) = A\ \mathrm{Cov}(X)\, A^T.$$

- The covariance matrix is always symmetric positive semidefinite. Geometrically, it records the spread of the cloud of points in each direction.
- A random vector $X \in \mathbb{R}^d$ is multivariate normal, written $X \sim \mathcal{N}_d(\mu, \Sigma)$, if every linear combination $u^T X$ is univariate normal. The vector μ is the center of the cloud and Σ controls its shape and orientation.
- If Σ is invertible, the multivariate normal density is

$$f_X(x) = \frac{1}{(2\pi)^{d/2} \det(\Sigma)^{1/2}} \exp\left(-\frac{1}{2}(x - \mu)^T \Sigma^{-1} (x - \mu) \right).$$

  Equal-density contours are ellipsoids, so correlation literally tilts and stretches the level sets.
- Gaussian vectors are stable under affine maps: if $X \sim \mathcal{N}_d(\mu, \Sigma)$ and $Y = BX + b$, then $Y \sim \mathcal{N}_m(B\mu + b, B\Sigma B^T)$. Marginals and conditionals are again Gaussian, and for jointly Gaussian coordinates, zero covariance is equivalent to independence.
- Useful distributions to know cold:
  - Binomial(n, p): counts the number of successes in a fixed number of independent Bernoulli trials. This is the natural law for repeated yes/no experiments with fixed horizon. Mean np, variance $np(1 - p)$.
  - Hypergeometric(N, K, n): counts marked objects in sampling *without* replacement. It is the finite-population version of the binomial, and the dependence comes from depletion of the population. Mean nK/N, variance

$$n\frac{K}{N}\left(1 - \frac{K}{N}\right)\frac{N - n}{N - 1}.$$

  - Poisson(λ): counts the number of rare, approximately independent arrivals in a time window or spatial region. It is the natural counting law for a constant-rate arrival process, and its mean equals its variance:

$$\mathbb{P}(X = k) = e^{-\lambda}\frac{\lambda^k}{k!}, \qquad \mathbb{E}[X] = \mathrm{Var}(X) = \lambda.$$

  - Geometric(p): waiting time until the first success in repeated Bernoulli trials. It is the discrete memoryless law:

$$\mathbb{P}(X = k) = (1 - p)^{k-1}p, \qquad \mathbb{E}[X] = \frac{1}{p}, \qquad \mathrm{Var}(X) = \frac{1 - p}{p^2}.$$

– Exponential(λ): continuous waiting time for the next event in a Poisson process. It is the continuous memoryless law, with density $\lambda e^{-\lambda x}$ on $x \geq 0$, mean $1/\lambda$, and variance $1/\lambda^2$.

– Uniform(a, b): every point in the interval is equally likely. This is the basic bounded-support model when there is no preferred location inside the interval. Mean $(a+b)/2$, variance $(b-a)^2/12$.

– Normal(μ, σ^2): the canonical law for aggregated noise and the CLT limit of many small independent shocks. It is symmetric, light-tailed, and fully determined by mean and variance. Its density is

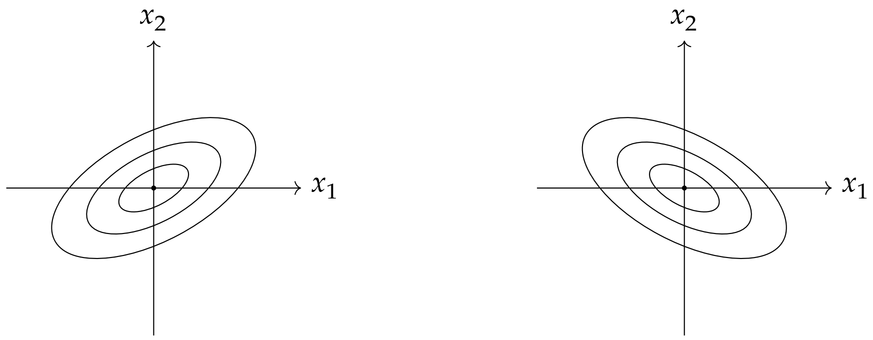
$$\frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right), \qquad \mathbb{E}[X] = \mu, \qquad \mathrm{Var}(X) = \sigma^2.$$

– Gamma(α, β) in the rate parameterization: a flexible positive-valued law for waiting times and intensities. When α is an integer, it is the sum of α i.i.d. exponential variables. It is useful when the object being modeled must stay positive and skewed. Mean α/β, variance α/β^2.

– Beta(a, b): a distribution on $[0, 1]$ used for random probabilities and proportions. Its shape can be U-shaped, symmetric, or strongly concentrated depending on a and b, which makes it useful in Bayesian updating for success probabilities. Mean $a/(a+b)$, variance $ab/[(a+b)^2(a+b+1)]$.

– Multivariate normal $\mathcal{N}_d(\mu, \Sigma)$: the vector-valued Gaussian law. It is determined by its mean vector μ and covariance matrix Σ, and its level sets are ellipsoids. This law appears constantly in factor models, linear signal models, and Gaussian portfolio calculations.

Contour sketch for a bivariate normal. The sign of the covariance changes the tilt of the ellipses while the eigenvalues of Σ control their axis lengths.



positive covariance   negative covariance

Example 4.4 (Choosing the right distribution)**.** Orders arrive according to a Poisson process with rate 3 per minute. What is the probability of exactly 5 orders in the next 2 minutes, and what is the probability that the next order arrives after 30 seconds?

Solution. If arrivals form a Poisson process with rate 3 per minute, then the number of arrivals in a 2-minute window has Poisson distribution with parameter

$$\lambda = 3 \times 2 = 6.$$

Therefore

$$\mathbb{P}(N = 5) = e^{-6}\frac{6^5}{5!}.$$

The waiting time T until the next arrival is exponential with rate 3 per minute. Since 30 seconds is 0.5 minutes,

$$\mathbb{P}(T > 0.5) = e^{-3 \cdot 0.5} = e^{-1.5}.$$

This example is useful because it highlights the meaning of two closely related laws: Poisson for counts in a window and exponential for waiting times between arrivals. $\square$

Example 4.5 (Broken stick triangle probability)**.** Break a stick of length 1 at two independent uniformly random points. What is the probability that the three pieces can form a triangle?

Solution. Let the breakpoints be $0 < U < V < 1$. The three lengths are

$$U, \qquad V - U, \qquad 1 - V.$$

They form a triangle if and only if no piece has length at least $1/2$. Thus we need

$$U < \frac{1}{2}, \qquad V - U < \frac{1}{2}, \qquad 1 - V < \frac{1}{2}.$$

Equivalently,

$$U < \frac{1}{2}, \qquad V > \frac{1}{2}, \qquad V < U + \frac{1}{2}.$$

The sample space is the triangle $0 < U < V < 1$, which has area $1/2$. The favorable region is a smaller central triangle with area $1/8$. Therefore

$$\mathbb{P}(\text{triangle}) = \frac{1/8}{1/2} = \frac{1}{4}.$$

So the answer is

$$\boxed{\frac{1}{4}}.$$

$\square$

Example 4.6 (Expected draw time for an inclusive marriage)**.** An inclusive marriage is a king together with a queen, with suits arbitrary. A standard deck is thoroughly shuffled, and cards are drawn one by one without replacement. Find the expected number of cards that must be drawn until an inclusive marriage has appeared. Express the answer as an irreducible fraction.

Solution. Let T be the number of cards drawn until we have seen at least one king and at least one queen.

Only the 8 special cards matter: the 4 kings and the 4 queens. Ignore the other 44 cards for a moment and look only at the special cards in the order they appear. This order is a uniformly random permutation of

$$K, K, K, K, Q, Q, Q, Q.$$

Let J be the index of the special card at which the first opposite rank appears. Then J can be $2, 3, 4$, or 5.

Its distribution is easy to compute:

$$\mathbb{P}(J = 2) = \frac{4}{7},$$

because after the first special card, 4 of the remaining 7 special cards have the opposite rank. Also,

$$\mathbb{P}(J = 3) = \frac{3}{7} \cdot \frac{4}{6} = \frac{2}{7},$$

$$\mathbb{P}(J=4) = \frac{3}{7}\cdot\frac{2}{6}\cdot\frac{4}{5} = \frac{4}{35},$$

and

$$\mathbb{P}(J=5) = \frac{3}{7}\cdot\frac{2}{6}\cdot\frac{1}{5} = \frac{1}{35}.$$

Therefore

$$\mathbb{E}[J] = 2\cdot\frac{4}{7} + 3\cdot\frac{2}{7} + 4\cdot\frac{4}{35} + 5\cdot\frac{1}{35} = \frac{13}{5}.$$

Now let S_j be the position in the full deck of the jth special card. The 44 non-special cards are split into 9 gaps around the 8 special cards, and by symmetry each gap has expected size $44/9$. Hence before the jth special card there are, on average, $j\cdot 44/9$ ordinary cards, so

$$\mathbb{E}[S_j] = j + \frac{44j}{9} = \frac{53j}{9}.$$

The positions of the 8 special cards are independent of which of those positions are occupied by kings and which by queens, so J is independent of the sequence (S_j). Since $T = S_J$, we obtain

$$\mathbb{E}[T] = \mathbb{E}[S_J] = \frac{53}{9}\mathbb{E}[J] = \frac{53}{9}\cdot\frac{13}{5} = \frac{689}{45}.$$

Thus the expected number of cards needed is

$$\boxed{\frac{689}{45}}.$$

□

Example 4.7 (Linear combination of a bivariate normal)**.** Let

$$X = \begin{pmatrix} X_1 \\ X_2 \end{pmatrix} \sim \mathcal{N}_2\left(\begin{pmatrix} 1 \\ -2 \end{pmatrix}, \begin{pmatrix} 4 & 1 \\ 1 & 9 \end{pmatrix}\right).$$

Find the distribution of $Y = 2X_1 - X_2$ and estimate $\mathbb{P}(Y > 0)$.

Solution. Write

$$a = \begin{pmatrix} 2 \\ -1 \end{pmatrix}.$$

Then $Y = a^T X$, and every linear combination of a multivariate normal vector is again normal. Therefore

$$Y \sim \mathcal{N}(a^T\mu, a^T\Sigma a).$$

Compute the mean:

$$a^T\mu = 2\cdot 1 - (-2) = 4.$$

Compute the variance:

$$a^T\Sigma a = \begin{pmatrix} 2 & -1 \end{pmatrix}\begin{pmatrix} 4 & 1 \\ 1 & 9 \end{pmatrix}\begin{pmatrix} 2 \\ -1 \end{pmatrix} = 21.$$

Hence

$$Y \sim \mathcal{N}(4, 21).$$

Standardizing gives

$$\mathbb{P}(Y > 0) = \mathbb{P}\left(Z > \frac{0-4}{\sqrt{21}}\right) = \Phi\left(\frac{4}{\sqrt{21}}\right) \approx 0.809,$$

where $Z \sim \mathcal{N}(0,1)$.

So the final answer is

$$\boxed{Y \sim \mathcal{N}(4, 21), \qquad \mathbb{P}(Y > 0) \approx 0.809.}$$

□

Example 4.8 (Mean and variance of an urn under a replace-if-black rule)**.** A basket holds m black balls and n red balls. At each step a uniformly random ball is drawn; if black it is replaced by a red ball, if red it is put back unchanged. Find $\mathbb{E}[R_k]$ and $\mathrm{Var}(R_k)$, the mean and variance of the number of red balls after k steps, as closed-form expressions (no sums or ellipses).

Solution. Label the $N = m + n$ physical balls; balls never leave the basket, and only the m originally black ones can change color. At each step the drawn label is uniform on $\{1, \dots, N\}$ and independent of the history (returning a red ball does not change the distribution of future draws). So a black ball i has turned red by time k exactly when its label was hit at least once among k i.i.d. uniform draws.

Let X_i ($i = 1, \dots, m$) indicate that black ball i has been drawn by time k, and set $a = \left(\frac{N-1}{N}\right)^k$, so $\mathbb{P}(X_i = 1) = 1 - a$. Since $R_k = n + \sum_{i=1}^{m} X_i$,

$$\mathbb{E}[R_k] = \boxed{n + m\left(1 - \left(\frac{N-1}{N}\right)^k\right)}, \qquad N = m + n.$$

For the variance, let $b = \left(\frac{N-2}{N}\right)^k$ be the probability that two fixed labels are *both* never drawn. By inclusion–exclusion, $\mathbb{P}(X_i = 1, X_j = 1) = 1 - 2a + b$ for $i \neq j$, so

$$\mathrm{Cov}(X_i, X_j) = (1 - 2a + b) - (1 - a)^2 = b - a^2,$$

while $\mathrm{Var}(X_i) = a(1 - a)$. Summing the m variances and $m(m-1)$ covariance pairs,

$$\mathrm{Var}(R_k) = ma(1 - a) + m(m-1)(b - a^2) = ma + m(m-1)b - m^2a^2,$$

that is,

$$\boxed{\mathrm{Var}(R_k) = m\left(\frac{N-1}{N}\right)^k + m(m-1)\left(\frac{N-2}{N}\right)^k - m^2\left(\frac{N-1}{N}\right)^{2k}}, \qquad N = m + n.$$

□

Example 4.9 (Expected value of the maximum of two independent exponentials)**.** $X \sim \mathrm{Exp}(1)$ and $Y \sim \mathrm{Exp}(2)$ are independent. Let $Z = \max(X, Y)$. Find $\mathbb{E}[Z]$.

Solution. Use $\max(X, Y) = X + Y - \min(X, Y)$, so $\mathbb{E}[Z] = \mathbb{E}[X] + \mathbb{E}[Y] - \mathbb{E}[\min(X, Y)]$. For independent exponentials with rates λ_1, λ_2, $\min(X, Y) \sim \mathrm{Exp}(\lambda_1 + \lambda_2)$ (the minimum of independent exponential "alarm clocks" rings at the combined rate). Here $\lambda_1 = 1, \lambda_2 = 2$, so $\min(X, Y) \sim \mathrm{Exp}(3)$, with mean $1/3$. Hence

$$\mathbb{E}[Z] = 1 + \frac{1}{2} - \frac{1}{3} = \boxed{\frac{7}{6}}.$$

□

4.3 Inequalities, Modes of Convergence, and Characteristic Functions

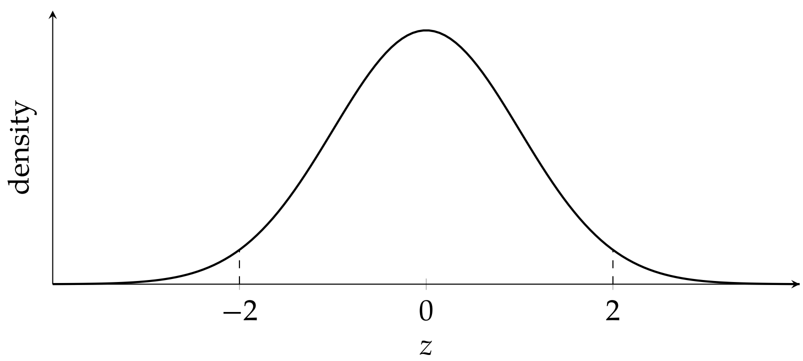
Definitions and formulas.

- Markov's inequality for nonnegative X:

$$\mathbb{P}(X \geq a) \leq \frac{\mathbb{E}[X]}{a}.$$

- Chebyshev's inequality:
$$\mathbb{P}(|X-\mu| \geq k\sigma) \leq \frac{1}{k^2}.$$
- Jensen's inequality: if φ is convex, then
$$\varphi(\mathbb{E}[X]) \leq \mathbb{E}[\varphi(X)].$$
- Borel–Cantelli: if $\sum_n \mathbb{P}(A_n) < \infty$, then only finitely many A_n occur almost surely.
- Common convergence notions:
  - almost sure: $X_n \to X$ a.s.;
  - in probability: $\mathbb{P}(|X_n - X| > \varepsilon) \to 0$;
  - in L^p: $\mathbb{E}[|X_n - X|^p] \to 0$;
  - in distribution: $F_{X_n}(x) \to F_X(x)$ at continuity points of F_X.
- The weak law says sample means converge in probability; the strong law says they converge almost surely; the central limit theorem describes the *fluctuation scale* around the law-of-large-numbers limit. For i.i.d. X_i with mean μ and variance $\sigma^2 < \infty$,
$$\sqrt{n}\,\frac{\bar{X}_n - \mu}{\sigma} \Rightarrow \mathcal{N}(0,1).$$
Equivalently, for large n,
$$\bar{X}_n \approx \mathcal{N}\left(\mu, \frac{\sigma^2}{n}\right), \qquad S_n = X_1 + \cdots + X_n \approx \mathcal{N}(n\mu, n\sigma^2).$$
For lattice variables such as binomial counts, a continuity correction often improves the approximation.
- The characteristic function of X is
$$\varphi_X(t) = \mathbb{E}[e^{itX}].$$
For independent X, Y, $\varphi_{X+Y}(t) = \varphi_X(t)\varphi_Y(t)$.

Standard normal reference curve. CLT approximations are usually read off after standardizing to the bell curve below.



Example 4.10 (Chebyshev bound for a sample mean)**.** Let $X_1, \dots, X_{400}$ be i.i.d. with mean 0 and variance 1. Give an upper bound on
$$\mathbb{P}\left(\left|\frac{1}{400}\sum_{i=1}^{400} X_i\right| > 0.1\right)$$
using Chebyshev's inequality.

Solution. Let $\bar{X}_{400} = \frac{1}{400}\sum_{i=1}^{400} X_i$. Then

$$\mathbb{E}[\bar{X}_{400}] = 0, \qquad \mathrm{Var}(\bar{X}_{400}) = \frac{1}{400^2}\cdot 400 = \frac{1}{400}.$$

By Chebyshev,

$$\mathbb{P}(|\bar{X}_{400}| > 0.1) \le \frac{\mathrm{Var}(\bar{X}_{400})}{0.1^2} = \frac{1/400}{0.01} = \frac{1}{4}.$$

Hence

$$\boxed{\mathbb{P}(|\bar{X}_{400}| > 0.1) \le \frac{1}{4}}.$$

This is conservative but completely distribution-free. □

Example 4.11 (Central limit approximation to a binomial count)**.** A fair coin is tossed 400 times. Estimate

$$\mathbb{P}(180 \le X \le 220),$$

where X is the number of heads.

Solution. Here

$$X \sim \mathrm{Bin}(400, 1/2), \qquad \mu = np = 200, \qquad \sigma = \sqrt{np(1-p)} = 10.$$

By the central limit theorem, X is well approximated by a normal variable with the same mean and variance. Using a continuity correction,

$$\mathbb{P}(180 \le X \le 220) \approx \mathbb{P}(179.5 \le Y \le 220.5),$$

where $Y \sim \mathcal{N}(200, 100)$.

Standardize:

$$\mathbb{P}(179.5 \le Y \le 220.5) = \mathbb{P}\left(\frac{179.5-200}{10} \le Z \le \frac{220.5-200}{10}\right) = \mathbb{P}(-2.05 \le Z \le 2.05),$$

where $Z \sim \mathcal{N}(0,1)$. Therefore

$$\mathbb{P}(180 \le X \le 220) \approx 2\Phi(2.05) - 1 \approx 0.9596.$$

So the CLT estimate is

$$\boxed{\mathbb{P}(180 \le X \le 220) \approx 0.960.}$$

□

Example 4.12 (A random increment process with shrinking step size)**.** Assume $x_3 > 0$ is fixed. For each $t \ge 3$, the process adds the current increment x_t to V, and then independently updates the next increment by

$$x_{t+1} = \begin{cases} x_t/2, & \text{with probability } 2/t, \\ x_t, & \text{with probability } 1 - 2/t. \end{cases}$$

What is the probability that $V_t \to \infty$ as $t \to \infty$?

Solution. Let I_t be the indicator that a halving occurs at time t, and let

$$N_n = \sum_{t=3}^{n} I_t$$

be the total number of halvings up to time n. Then

$$x_n = x_3\, 2^{-N_{n-1}}.$$

Since the halving decisions are independent and $\mathbb{P}(I_t = 1) = 2/t$,

$$\mathbb{E}[N_n] = \sum_{t=3}^{n} \frac{2}{t} = 2\log n + O(1).$$

By the strong law for sums of independent Bernoulli variables with varying parameters,

$$N_n = 2\log n + o(\log n) \qquad \text{almost surely.}$$

Therefore

$$x_n = x_3 \exp(-(\log 2)N_{n-1}) = n^{-2\log 2 + o(1)} \qquad \text{almost surely.}$$

Now

$$2\log 2 \approx 1.386 > 1,$$

so almost surely the series $\sum_n x_n$ behaves like a p-series with exponent strictly larger than 1. Hence

$$\sum_{n=3}^{\infty} x_n < \infty \qquad \text{almost surely.}$$

But V_t is obtained by summing these increments, so V_t converges to a finite random limit almost surely. Consequently,

$$\boxed{\mathbb{P}(V_t \to \infty) = 0}.$$

This is a useful warning example: even though the expected number of halvings by time n is only logarithmic, that is still enough to make the typical increment decay faster than $1/n$, so the cumulative growth stays finite almost surely. □

4.4 Markov Chains and First-Step Analysis

Definitions and formulas.

- A Markov chain has transition matrix $P = (p_{ij})$, where

  $$p_{ij} = \mathbb{P}(X_{n+1} = j \mid X_n = i).$$

  Each row of P sums to 1. The key modeling step is to choose the state so that, once the current state is known, the future no longer depends on the earlier history. In interview problems, the hard part is usually not matrix algebra but selecting the right state description.

- A distribution π is stationary if $\pi P = \pi$. This means that if the chain starts with distribution π, then after one step it still has distribution π. For ergodic finite chains, this is the long-run equilibrium law.

- In a finite irreducible and aperiodic chain, the stationary distribution is unique and the chain converges to it regardless of the starting state. Irreducible means every state can eventually reach every other state, and aperiodic means the chain is not trapped in a rigid cycle length.

- Hitting probabilities are handled by the same one-step idea. If

  $$q_i = \mathbb{P}_i(\tau_A < \tau_B),$$

  then the boundary conditions are

  $$q_i = 1 \text{ for } i \in A, \qquad q_i = 0 \text{ for } i \in B,$$

and for all other states,

$$q_i = \sum_j p_{ij} q_j.$$

There is no extra 1 here because we are computing a probability, not a time. We simply average the future success probability over the next state.

- Expected hitting times are often solved by *first-step analysis*: condition on the next move and write a recursion.
- If τ_A is the first hitting time of a set A, then the expected hitting times satisfy

$$h_i = 0 \text{ for } i \in A, \qquad h_i = 1 + \sum_j p_{ij} h_j \text{ for } i \notin A.$$

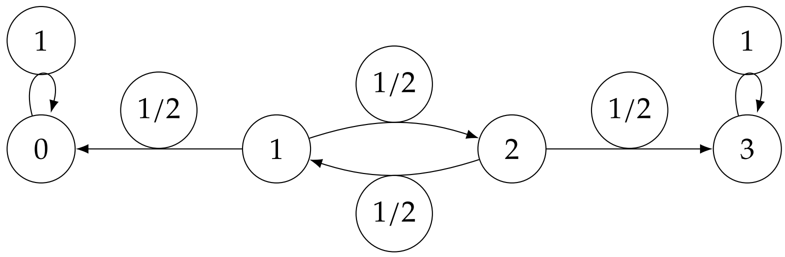
The extra 1 is the step you take immediately before landing in the next state.

- Absorbing-state problems reduce to linear systems. Once you assign one variable to each transient state, the recursion from first-step analysis produces one linear equation per unknown.
- For an absorbing Markov chain with block form

$$P = \begin{pmatrix} Q & R \\ 0 & I \end{pmatrix},$$

the matrix Q describes transitions among transient states, R describes jumps from transient states to absorbing states, and the fundamental matrix is $N = (I - Q)^{-1}$. The entry N_{ij} is the expected number of visits to transient state j when starting from transient state i, before absorption occurs. Summing across a row gives the expected absorption time, so $N\mathbf{1}$ gives expected absorption times from transient states.

Transition-graph picture. The matrix and the directed graph contain the same information: draw one node per state and an arrow $i \to j$ labeled by p_{ij} whenever $p_{ij} > 0$. For the random walk example below, the chain can be visualized as follows.



The absorbing states 0 and 3 have self-loops of probability 1. States 1 and 2 are transient because, once the walk enters 0 or 3, it never returns.

Practical workflow.

1. Choose states that retain exactly the information that still matters for the future.
2. Write one equation per nonterminal state by conditioning on the next step.
3. Put boundary values on terminal or absorbing states.
4. Solve the resulting linear system.
5. Read off the quantity corresponding to the actual starting state.

In interview practice there are three recurring templates:

- If you are asked for a probability of eventually hitting one target before another, define q_i and use $q_i = \sum_j p_{ij} q_j$.
- If you are asked for an expected waiting time, define h_i and use $h_i = 1 + \sum_j p_{ij} h_j$.
- If the process only depends on a small memory of the past, compress the past into a few states. The two-sixes example below is exactly of this type.

Example 4.13 (Expected rolls until two consecutive sixes)**.** You roll a fair six-sided die repeatedly. What is the expected number of rolls until two consecutive sixes appear?

Solution. Use two states, chosen so that the future depends only on whether we are currently carrying a trailing 6:

- E_0: expected additional rolls when we do *not* currently have a trailing 6;
- E_1: expected additional rolls when the previous roll was a 6, so one more 6 finishes the game.

At the start of the experiment no roll has happened yet, so we are in state E_0, not E_1. That is why the answer to the problem will be E_0.

This is the standard state-compression trick: the whole past sequence of rolls is irrelevant except for one bit of information, namely whether the most recent roll was a 6.

From state E_0, one roll occurs, and then with probability 5/6 we remain in E_0, while with probability 1/6 we move to E_1:

$$E_0 = 1 + \frac{5}{6}E_0 + \frac{1}{6}E_1.$$

From state E_1, one roll occurs. With probability 1/6 we finish, and with probability 5/6 we return to E_0:

$$E_1 = 1 + \frac{5}{6}E_0.$$

The first equation simplifies to

$$E_0 = 6 + E_1.$$

Substitute into the second:

$$E_1 = 1 + \frac{5}{6}(6 + E_1) = 6 + \frac{5}{6}E_1.$$

Thus

$$\frac{1}{6}E_1 = 6 \quad \Longrightarrow \quad E_1 = 36,$$

and therefore

$$E_0 = 6 + 36 = 42.$$

Because the chain starts in state E_0, the expected number of rolls asked for in the problem is

$$\boxed{42}.$$

The structure is worth remembering: one equation per nonterminal state, one immediate step cost, and then an average over the next states. □

Example 4.14 (Absorption on a short random walk)**.** A particle moves on the states 0, 1, 2, 3. States 0 and 3 are absorbing. From states 1 and 2, the particle moves left or right with probability 1/2 each. If the chain starts at state 1, find the probability of being absorbed at 3 and the expected time to absorption.

Solution. Let p_i be the probability, starting from state i, of hitting 3 before 0. Then

$$p_0 = 0, \qquad p_3 = 1,$$

and by first-step analysis,

$$p_1 = \frac{p_0 + p_2}{2} = \frac{p_2}{2}, \qquad p_2 = \frac{p_1 + p_3}{2} = \frac{p_1 + 1}{2}.$$

Substituting the first equation into the second gives

$$p_2 = \frac{p_2/2 + 1}{2},$$

so $4p_2 = p_2 + 2$, hence $p_2 = \frac{2}{3}$ and therefore

$$p_1 = \frac{1}{3}.$$

Thus the probability of absorption at 3 when starting from 1 is

$$\boxed{\frac{1}{3}}.$$

This is the hitting-probability template in its purest form: no extra 1 appears because we are not counting time, only eventual success.

Now let e_i be the expected time to absorption starting from state i. Then

$$e_0 = e_3 = 0,$$

and

$$e_1 = 1 + \frac{e_0 + e_2}{2} = 1 + \frac{e_2}{2}, \qquad e_2 = 1 + \frac{e_1 + e_3}{2} = 1 + \frac{e_1}{2}.$$

Substitute the second equation into the first:

$$e_1 = 1 + \frac{1 + e_1/2}{2} = \frac{3}{2} + \frac{e_1}{4}.$$

Hence

$$\frac{3e_1}{4} = \frac{3}{2}, \qquad e_1 = 2.$$

So the expected time to absorption is

$$\boxed{2}.$$

Here the extra 1 appears because one step is spent immediately, regardless of which direction the walk takes next. □

5 Algorithm Design

Quant interviews frequently include short algorithmic problems that reward recognizing a classical pattern — a monotonic stack, a sliding window, a divide-and-conquer trick — over brute force.

Example 5.1 (Nearest right element at least double, in $O(n \log n)$)**.** Given a real array $A[1..n]$, design an algorithm that, for every index i, finds the smallest index $j > i$ with $A[j] \geq 2A[i]$ (or reports that none exists), using $O(n \log n)$ time and $O(n)$ extra space.

Solution. **A domination principle.** Scan the array from right to left, maintaining a stack of surviving candidates. Say index j_2 is *dominated* by a nearer index $j_1 < j_2$ if $A[j_1] \geq A[j_2]$. A dominated candidate is useless: for any future (further left) query i, whenever $A[j_2] \geq 2A[i]$ we also have $A[j_1] \geq A[j_2] \geq 2A[i]$, and j_1 is strictly closer, so j_2 can never be the nearest qualifying index once j_1 exists. Notably, this domination argument does not depend on the multiplier 2; it only uses $A[j_1] \geq A[j_2]$.

Maintaining the stack. Process $i = n, n-1, \ldots, 1$. Before pushing i, pop every index currently on top of the stack with value $\leq A[i]$ (those are now dominated by i), then push i. Each index is pushed and popped at most once, so stack maintenance costs $O(n)$ total.

Why the stack is sorted. Whenever an element is pushed, everything weaker has just been popped, so values strictly increase from the bottom of the stack toward the top — equivalently, reading from the top (nearest to the current scan position) down to the bottom (farthest), values are non-decreasing. So at every moment the live stack is a sequence sorted by value, in which position along the stack corresponds monotonically to distance from the current pointer.

Answering a query. For index i we want the surviving candidate closest to the top with value $\geq 2A[i]$. Because the stack is sorted by value from top (smallest) to bottom (largest), the qualifying candidates form a contiguous block at the bottom, and the nearest one is found by *binary search* on the stack (kept as an array) for the threshold $2A[i]$: $O(\log n)$ per query.

Complexity. Maintenance is $O(n)$ amortized; each of the n queries costs $O(\log n)$ via binary search on the array-backed stack; total time is $O(n \log n)$, and the stack never holds more than n indices, so space is $O(n)$ — matching the required bounds exactly. (A plain monotonic stack alone, without the binary search, solves the classical "next strictly greater element" problem in $O(n)$; the extra $\log n$ factor here is exactly the cost of the per-query threshold search that the "at least double" condition forces.) □

How to Use This Guide Efficiently

- Memorize formulas only after you understand where they come from.
- When a problem looks hard, first classify it: parity, conditioning, asymptotics, optimization, or linear structure.
- In an interview, write down the state variables explicitly. A correct state definition is usually half of the solution.
- For revision, cover the solutions and try to reproduce them from the definitions and formulas alone.